EVIDENCE OVER ASSUMPTION

A practical guide for presenting the thesis through its public dashboard.

50 paintings
525 registered cases
410 restoration-eligible cases
1,785 approved candidates
8 evidence views

SUPERVISOR DASHBOARD WALKTHROUGH GUIDE

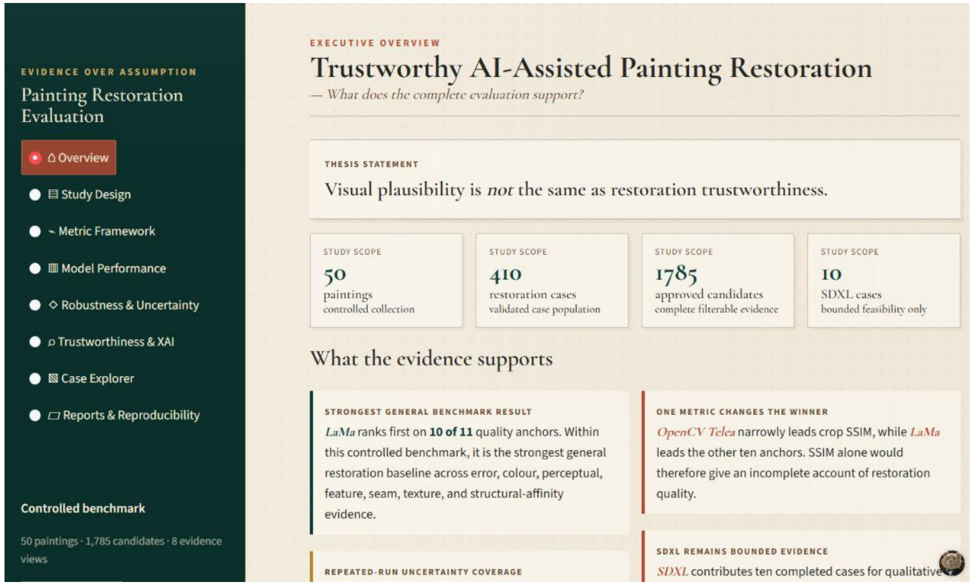
Trustworthy AI-Assisted Painting Restoration

How to navigate, explain, and defend the complete evaluation

Visual plausibility is not the same as restoration trustworthiness.

Open the live Streamlit dashboard

Prepared 2026-09-05 · Uses validated, saved evidence only



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How to use this guide

Use the dashboard as the primary story and the notebooks as the audit trail. The guide is structured so each screen answers four questions: what changed, what was measured, what the result means, and what the result cannot establish.

Route	Use when	Pages to prioritize	Outcome
10-minute overview	A short supervisor check-in	Overview → Model Performance → Case Explorer → Reports	One conclusion, one counterexample, one traceability path
25-minute methods tour	The evaluation design is the focus	Study Design → Metric Framework → Robustness → Case Explorer	How cases, regions, metrics, and uncertainty fit together
45-minute full defence rehearsal	A complete walkthrough	All eight tabs in sequence	A claim-evidence-limit narrative with numerical and visual examples

Visual cue	Meaning in this guide
Navigate / set	The dashboard control or filter to select before speaking.
Read	The exact visual or numerical evidence that deserves attention.
Supported conclusion	The strongest statement justified by the displayed evidence.
Do not claim	The interpretation boundary that preserves scientific honesty.
Evidence trail	The producing notebooks and saved artifacts to open if challenged.

Contents and live navigation

Live dashboard: https://fhtw-painting-restoration.streamlit.app/

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The thesis logic in one page

Question	Design response	Evidence retained	Interpretation boundary
Is the repair close to the reference?	Classical, perceptual, feature, colour, texture, seam, and structural evidence	Separate metric-region rows and maps	No single metric establishes trustworthiness
Does performance change under controlled variation?	Damage extent, mask placement, synthetic degradation, and prompt sensitivity	Matched trajectories and grouped summaries	Sensitivity does not diagnose the failure mechanism
Does the stochastic model agree with itself?	Four Stable Diffusion seeds, fixed case/prompt/configuration	165 complete groups and spatial variability maps	Agreement is not correctness; disagreement is not calibrated confidence
Can a reviewer trace a warning?	Rule-derived flags plus images, metrics, maps, and provenance	1,785 filterable candidates and 23,964 visual records	Flags are review guidance, not expert ground truth

Population	Count	How to say it
Paintings	50	Balanced across five broad visual categories; these are not validated art-historical styles.
Registered cases	525	250 canonical + 35 damage-size + 75 mask-robustness + 165 synthetic-degradation cases.
Restoration-eligible cases	410	All canonical, damage-size, and mask cases, plus 50 eligible masked-removal degradation cases.
Approved candidate records	1,785	410 LaMa + 410 Telea + 955 Stable Diffusion + 10 bounded SDXL candidates.

Research questions and defensible answers

RQ	Proposal wording	Evidence-based answer	Necessary qualification
RQ1	What additional evidence does a region-aware, multi-metric evaluation framework provide beyond traditional image-similarity metrics when evaluating AI-assisted painting restoration?	Separate metric families and spatial regions show reference, perceptual, feature, local, and spatial evidence that traditional similarity alone would hide.	No universal quality or trustworthiness score is claimed.
RQ2	How do selected inpainting methods differ in restoration quality across controlled artificial damage conditions, and how consistent are these differences across the evaluated paintings?	Across paired controlled damage conditions, LaMa is the strongest general baseline on 10 of 11 anchors; Telea narrowly leads crop SSIM; Stable Diffusion is more variable; SDXL remains bounded.	Painting-level and condition-specific evidence tests consistency; visual categories remain descriptive and are not independently estimated style effects.
RQ3	How can repeated-candidate disagreement be used to characterize the stability of stochastic painting restorations, and how does it relate to other restoration-quality diagnostics?	Repeated-candidate disagreement localizes where Stable Diffusion restorations are less stable and relates that stability to quality and spatial diagnostics.	Disagreement characterizes stability, not correctness or calibrated confidence; low variation can still be consistently wrong.

Presentation principle: state the result directly, then immediately state the population, region, metric, and limitation that make it defensible.

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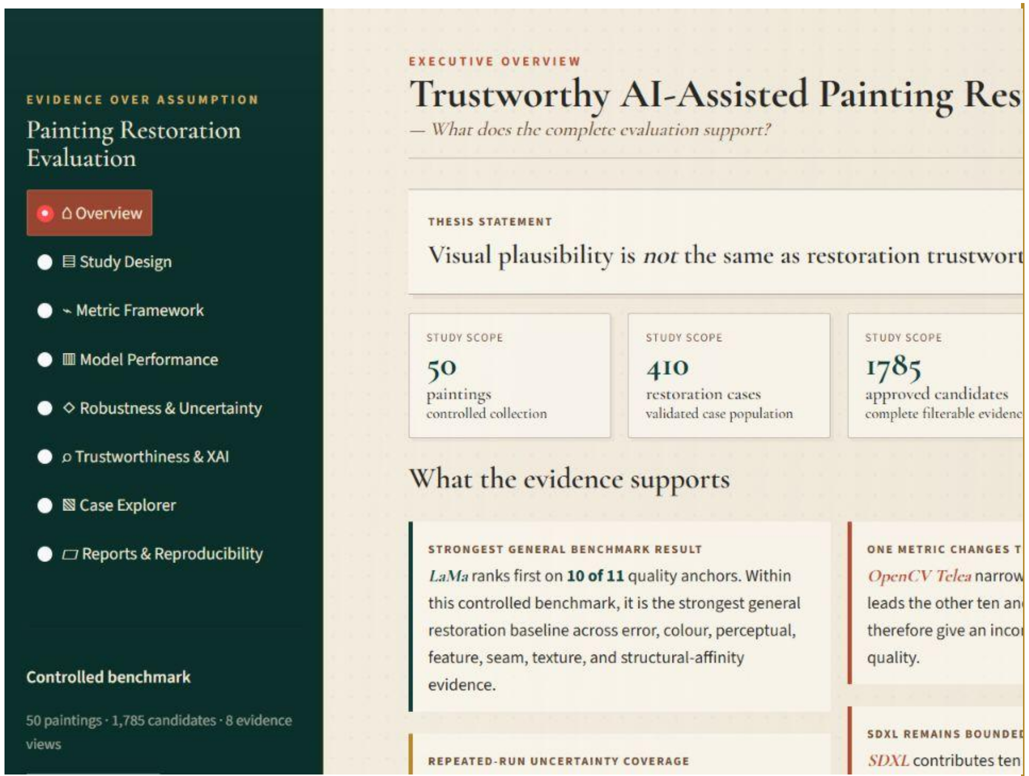
From painting to defensible conclusion

Stage	Notebooks	Responsibility	Inspectable result
1	N01–N08	Register paintings → preprocess → generate cases/masks → freeze eligibility and region policy	Stable IDs, 768×768 RGB images, content bounds, 525-case registry
2	N09–N12	Generate restoration candidates	Telea, LaMa, Stable Diffusion, bounded SDXL outputs
3	N13–N20, N22	Compute complementary metrics, maps, local consistency, semantics, and repeated-seed variability	Scalar rows, pairwise evidence, heatmaps, region-aware diagnostics
4	N21, N23–N28	Compare methods and test sensitivity, statistics, failures, and metric policy	Grouped estimates, intervals, effect sizes, review flags
5	N29–N33	Explain cases and generate reports	Retrieval context, model cards, 104 self-contained reports
6	N34–N36	Package dashboard, validate deployment, and produce supervisor/reproducibility assets	1,785 candidate index, 23,964 visual records, manifests, checksums, zero blocking N34 checks

The dashboard performs presentation and filtering only. It reads saved evidence; it does not run restoration inference or recompute metrics.

Orient the audience before discussing a model



Overview: thesis statement and validated scope

NAVIGATE / SET

Open Overview. Keep every filter at its default; this page is the fixed thesis-level entry point.

READ

Four scope cards separate paintings, restoration cases, approved candidates, and the bounded SDXL subset.

SUPPORTED CONCLUSION

The study is broad enough for controlled comparison while remaining explicit about the bounded SDXL branch.

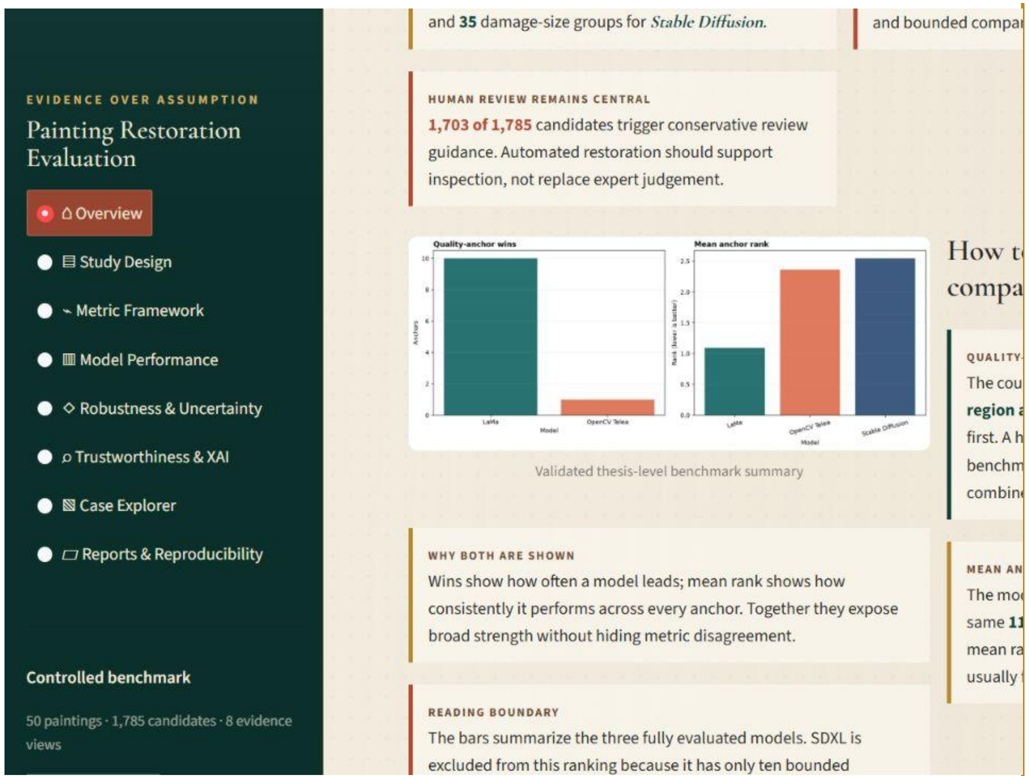
DO NOT CLAIM

The counts describe this benchmark, not all painting traditions or real conservation practice.

EVIDENCE TRAIL

N34 dashboard package; N36 supervisor package

Explain quality-anchor wins and mean anchor rank



Overview: benchmark bars and their reading guide

NAVIGATE / SET

Scroll to the two-bar summary and the 'How to read the comparison' text.

READ

Wins count first-place results across 11 metric-region anchors. Mean rank averages a model's position across those same anchors; lower is better.

SUPPORTED CONCLUSION

LaMa's 10 wins and near-first mean rank indicate broad benchmark leadership without collapsing metrics into one score.

DO NOT CLAIM

Anchor wins are descriptive. They are not a universal quality score, and SDXL is excluded because it has only ten bounded cases.

EVIDENCE TRAIL

N21 model comparison; N34 performance_summary.csv

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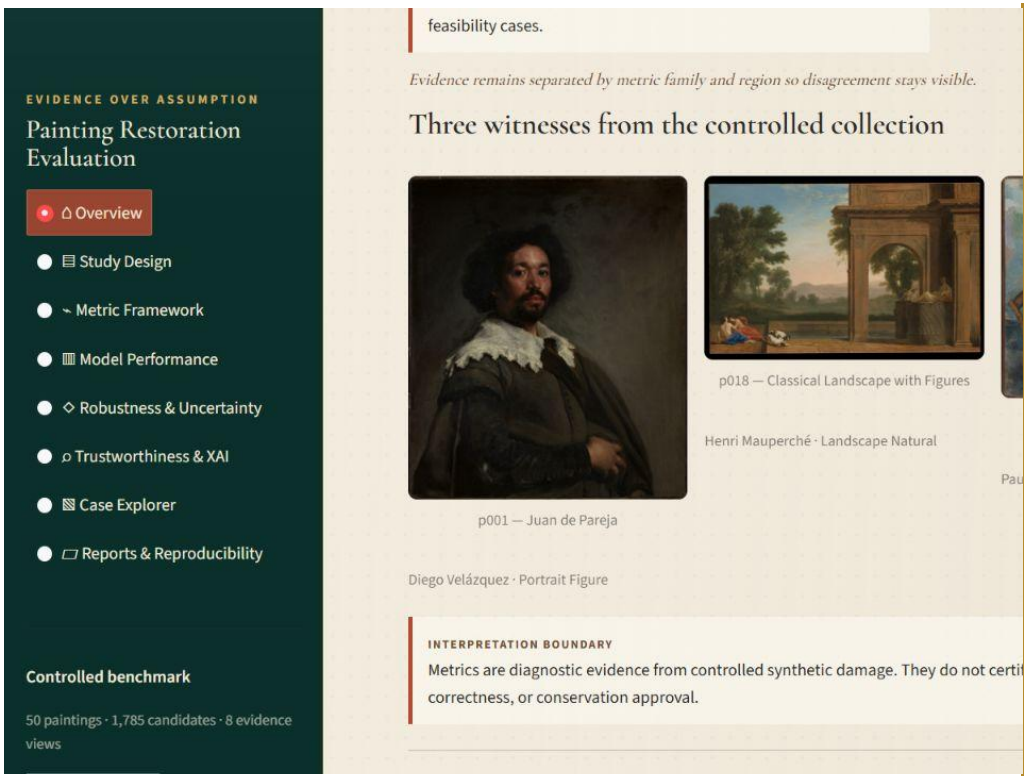
Use three paintings as witnesses, not decorations



Overview: representative paintings and the interpretation boundary

NAVIGATE / SET

Scroll to 'Three witnesses from the controlled collection'.

READ

p001 shows portrait/figure detail, p018 landscape structure, and p043 high-texture brushwork. Use them as recurring visual witnesses.

SUPPORTED CONCLUSION

A claim should be illustrated on more than one visual category whenever the dashboard offers the evidence.

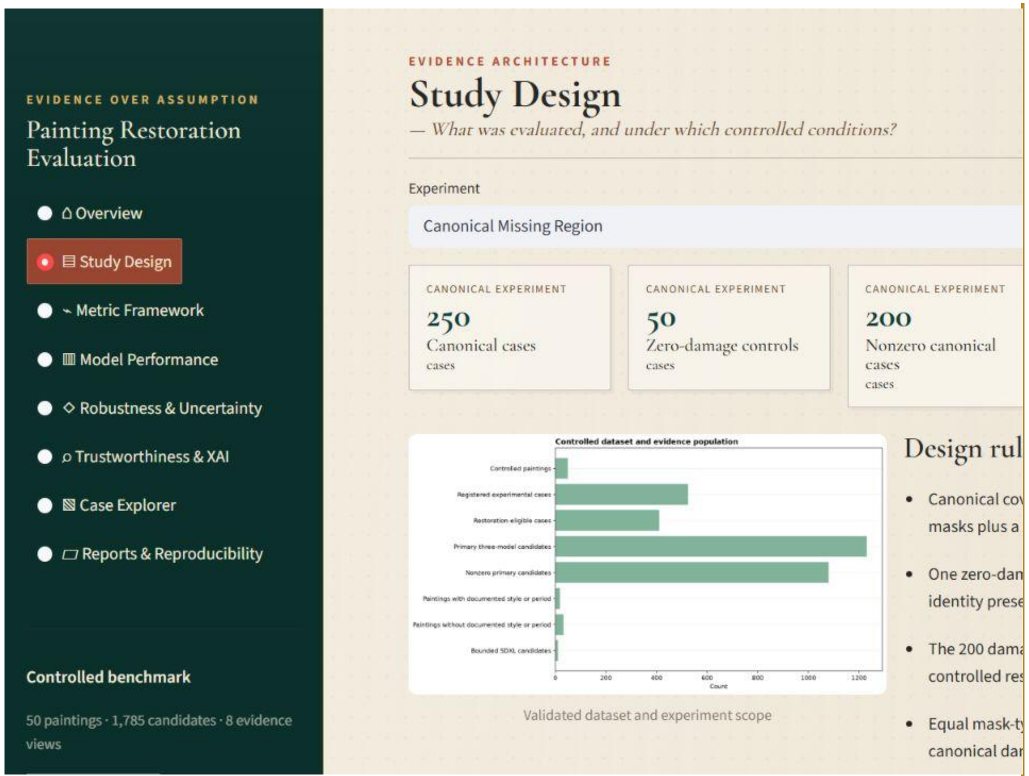
DO NOT CLAIM

Representative examples illustrate stored results; they do not replace the complete 50-painting tables.

EVIDENCE TRAIL

N02 clean images; N32 painting reports; N34 painting index

Start with the canonical experiment and independent unit



Study Design: canonical scope and design rules

NAVIGATE / SET

Open Study Design and choose Canonical Missing Region.

READ

Every painting receives four damage masks plus one zero-damage control: 250 cases total, 200 with nonzero damage.

SUPPORTED CONCLUSION

Equal mask-family coverage supports controlled within-study comparison; paintings, not candidate rows, are the independent unit.

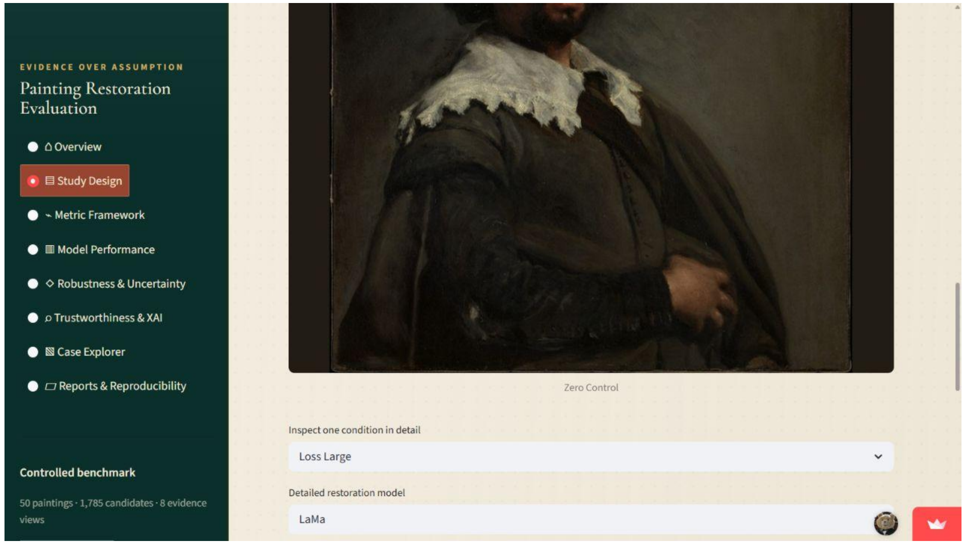
DO NOT CLAIM

Repeated seeds and multiple models are nested observations, not new independent paintings.

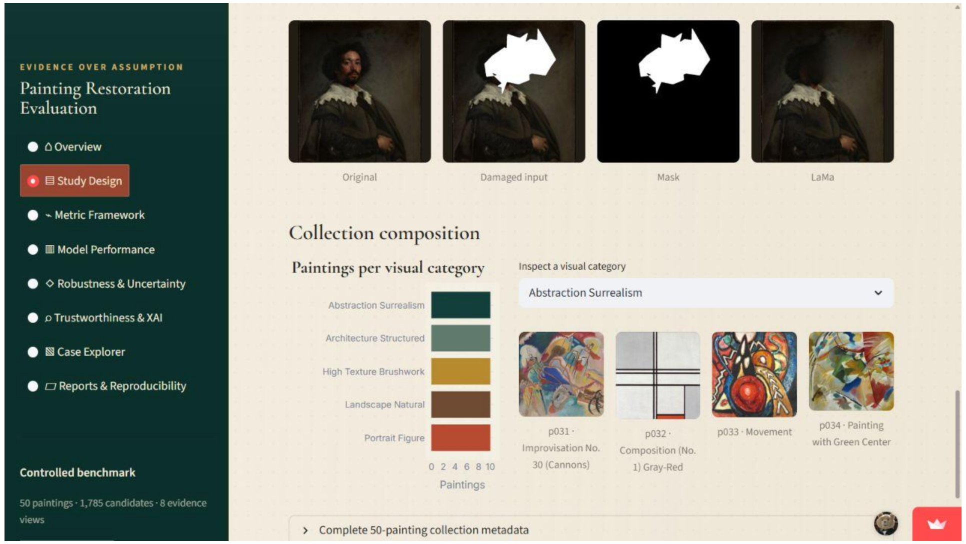
EVIDENCE TRAIL

N01–N04; N08 case registry

Show how a canonical case is constructed



Canonical damage gallery



Original, damaged input, mask, and LaMa restoration

READ

White mask pixels define the region the model may repair. The same case ID links every model output and diagnostic.

SUPPORTED CONCLUSION

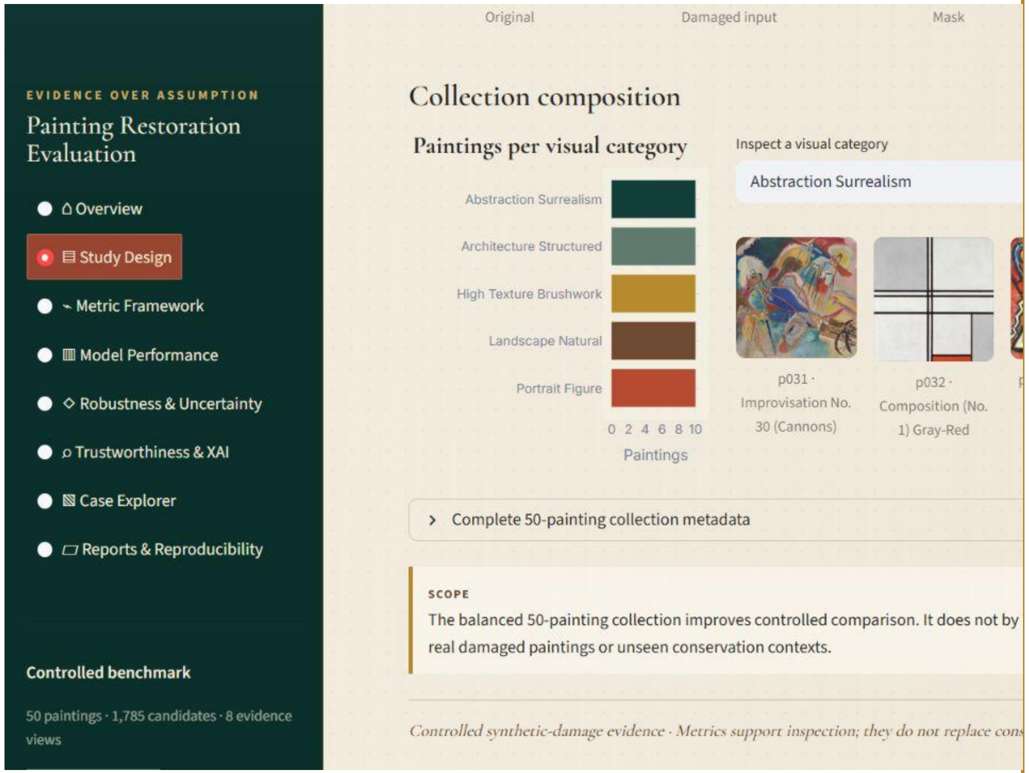
The comparison is reference-based reconstruction under synthetic missingness, not recovery of genuinely unknown historical content.

LIMIT

A visually plausible fill can still be structurally, texturally, or historically wrong.

Evidence trail: N03 masks; N04 damaged images; N09–N12 restorations

Explain category balance without calling it style validation



Study Design: ten paintings in each broad visual category

NAVIGATE / SET

Scroll to Collection composition and change 'Inspect a visual category'.

READ

The collection has ten paintings in each of five operational visual categories.

SUPPORTED CONCLUSION

Balanced categories prevent one broad visual type from dominating the benchmark.

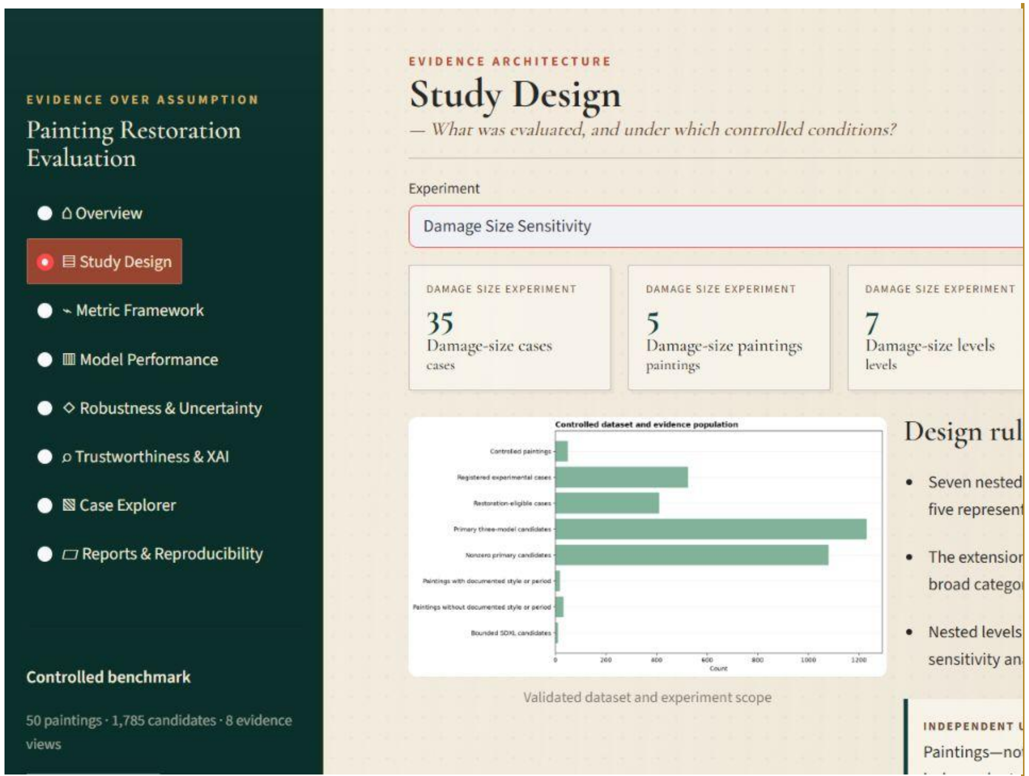
DO NOT CLAIM

The five categories are not validated art-historical styles; complete style/period metadata is unavailable for much of the collection.

EVIDENCE TRAIL

N01 verified dataset registry; N34 study_design.csv

Define the damage-size sensitivity experiment



Study Design: damage-size experiment summary

NAVIGATE / SET

Set Experiment to Damage Size Sensitivity.

READ

Five paintings are evaluated at seven nested target damage levels: 2%, 4%, 6%, 8%, 10%, 15%, and 20%.

SUPPORTED CONCLUSION

Holding painting and damage family fixed isolates sensitivity to the amount of missing content.

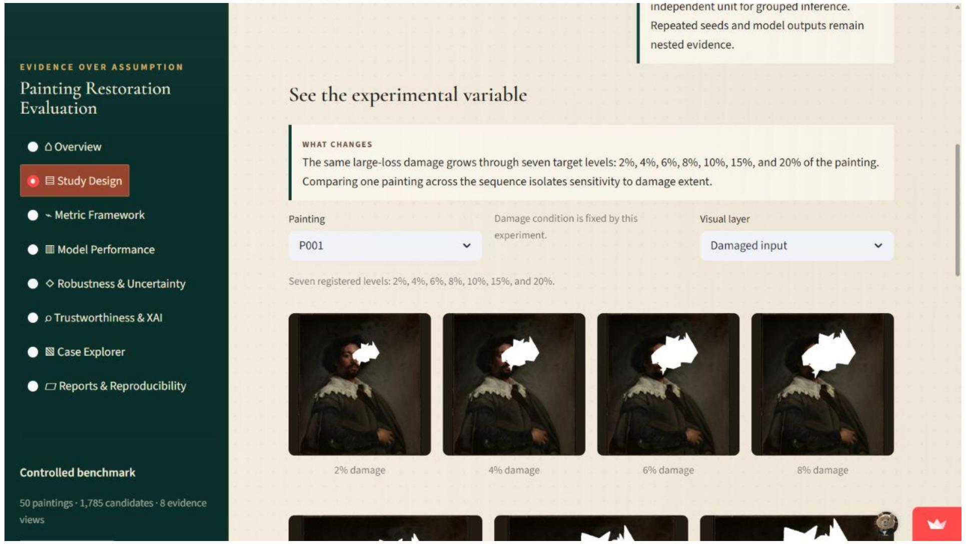
DO NOT CLAIM

The five-painting cohort cannot estimate an independent visual-category effect.

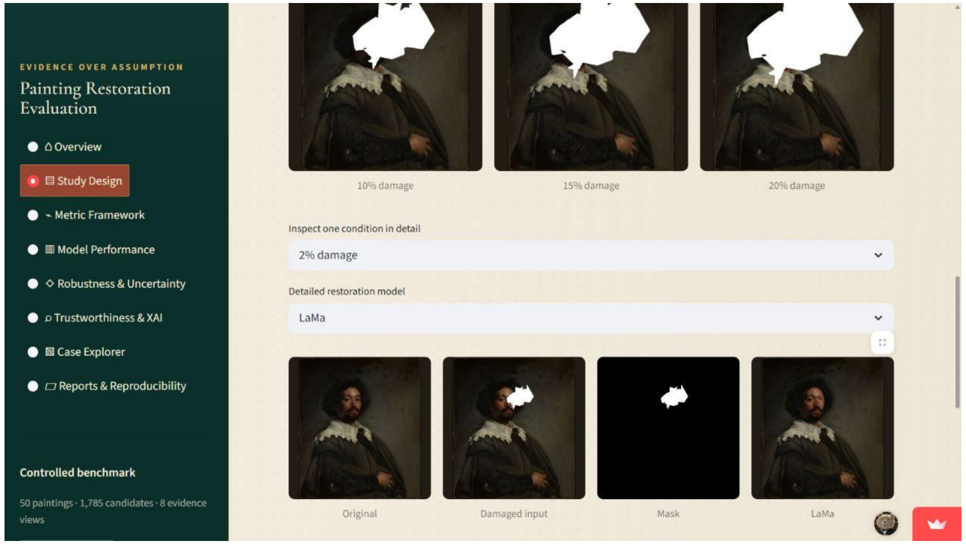
EVIDENCE TRAIL

N05 dataset; N22 Stable Diffusion extension; N23 analysis

Make damage extent visible before showing a slope



Seven visible damage levels



One selected level with original, damaged, mask, and restoration

READ

The sequence converts an abstract percentage into visible missing area. The detail row verifies that images and masks match the selected level.

SUPPORTED CONCLUSION

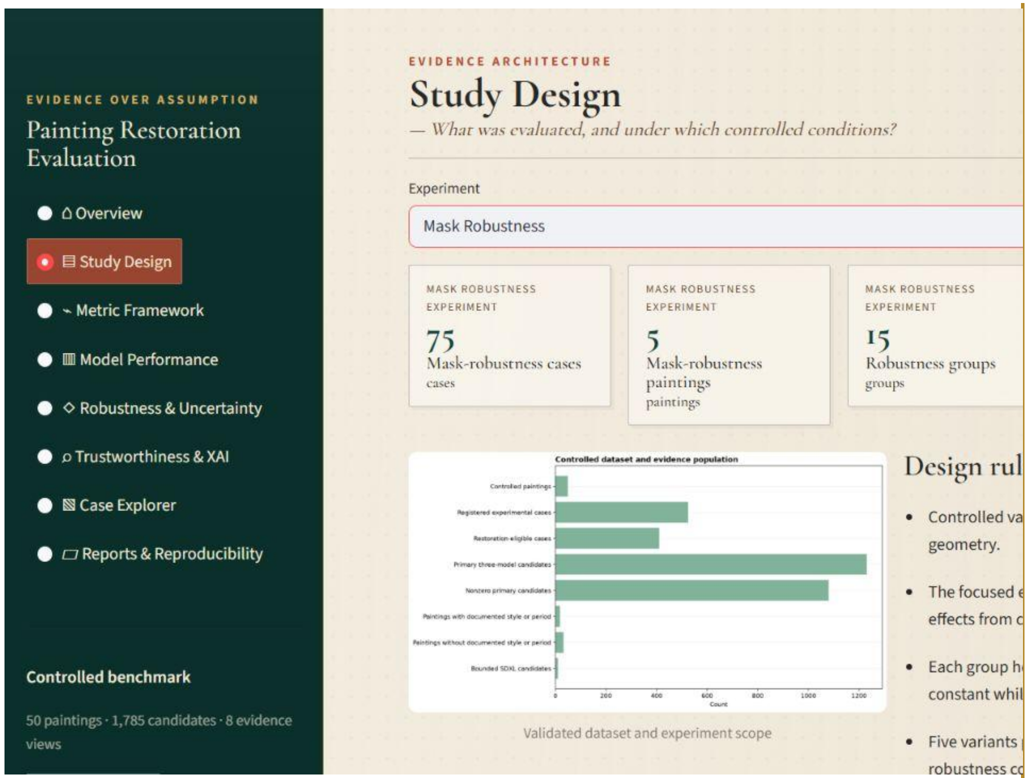
A steeper metric trajectory means stronger sensitivity to missing-area extent for that model and metric.

LIMIT

A percentage is not a severity judgement about real conservation damage.

Evidence trail: N05 figures and manifests; N23 trajectories

Define mask-placement robustness



Study Design: matched mask-placement experiment

NAVIGATE / SET

Set Experiment to Mask Robustness.

READ

Five placements are tested for each selected painting, damage type, and target size; the mask geometry and location change.

SUPPORTED CONCLUSION

Differences across placements measure robustness to where and how the missing region occurs.

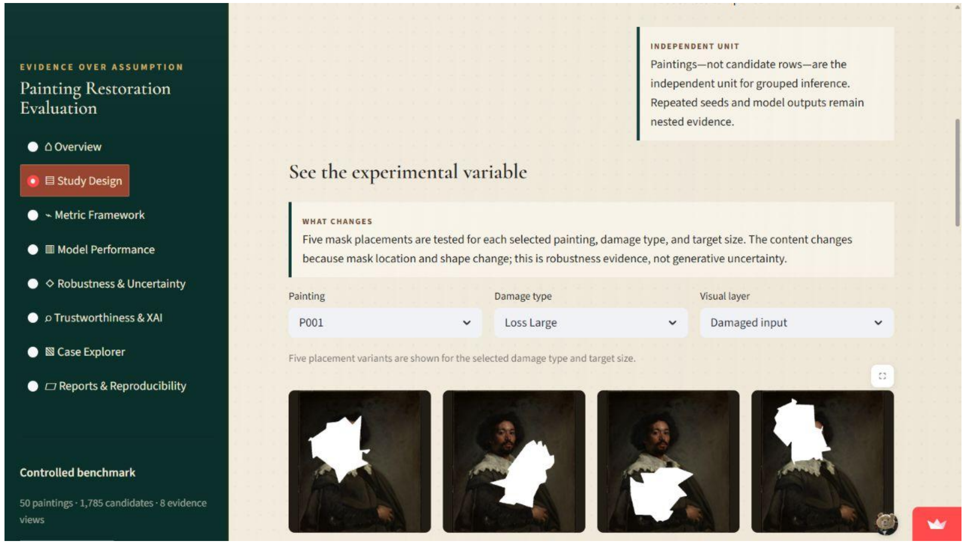
DO NOT CLAIM

Mask-placement variation is not stochastic uncertainty because no seed changes are involved.

EVIDENCE TRAIL

N06 dataset; N24 robustness analysis

Compare matched mask geometries visually



Mask-placement gallery

Selected placement with model result

READ

Compare the same painting and target size across placements, then inspect the chosen restoration at full resolution.

SUPPORTED CONCLUSION

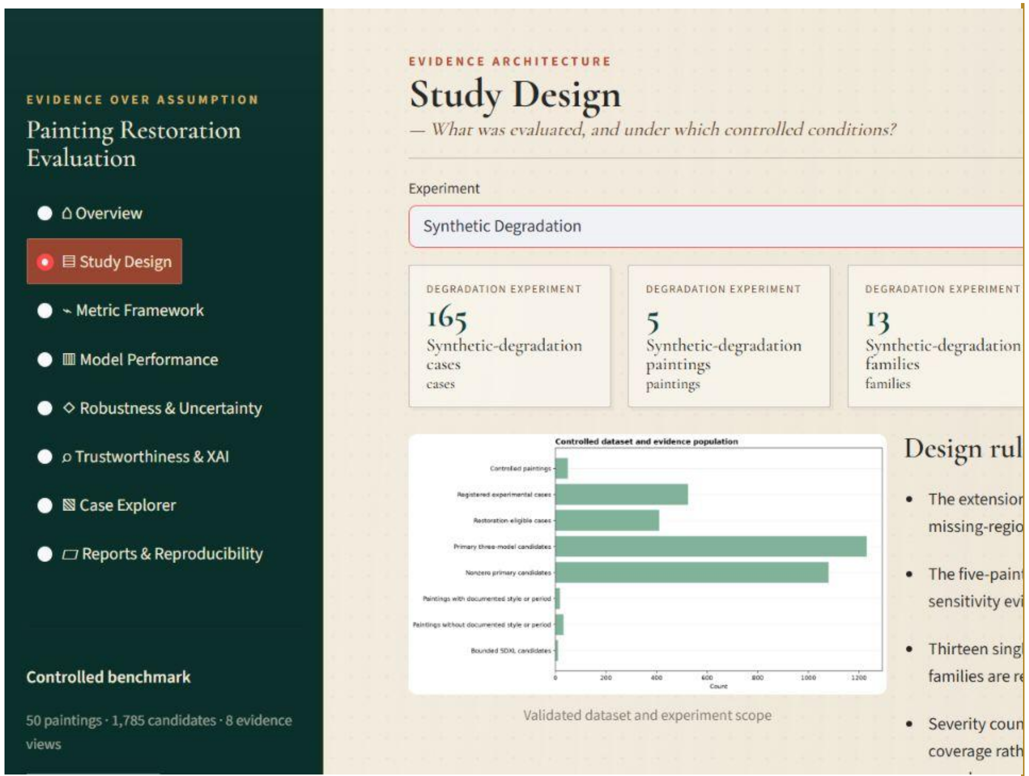
A method is more robust when its quality changes less across matched placements.

LIMIT

Five paintings still limit generalization; report descriptive or paired grouped evidence as designed.

Evidence trail: N06 images; N24 paired summaries

Separate broad degradation from missing-region restoration



Study Design: procedural synthetic degradation scope

NAVIGATE / SET

Set Experiment to Synthetic Degradation.

READ

The registry contains 165 cases across 13 single and combined degradation families on five paintings.

SUPPORTED CONCLUSION

This branch tests sensitivity beyond binary missing-region masks.

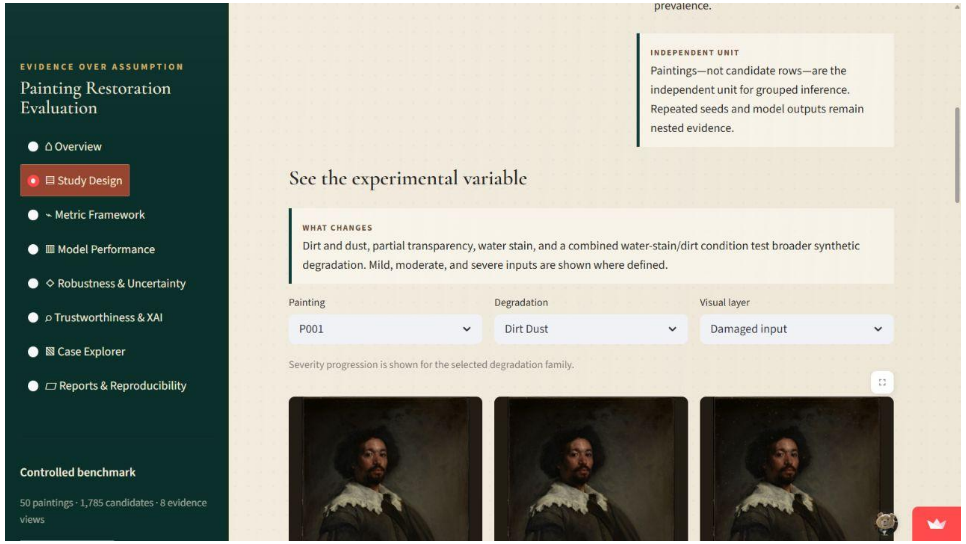
DO NOT CLAIM

Only 50 cases with mask-like removable effects enter the selected inpainting comparison.

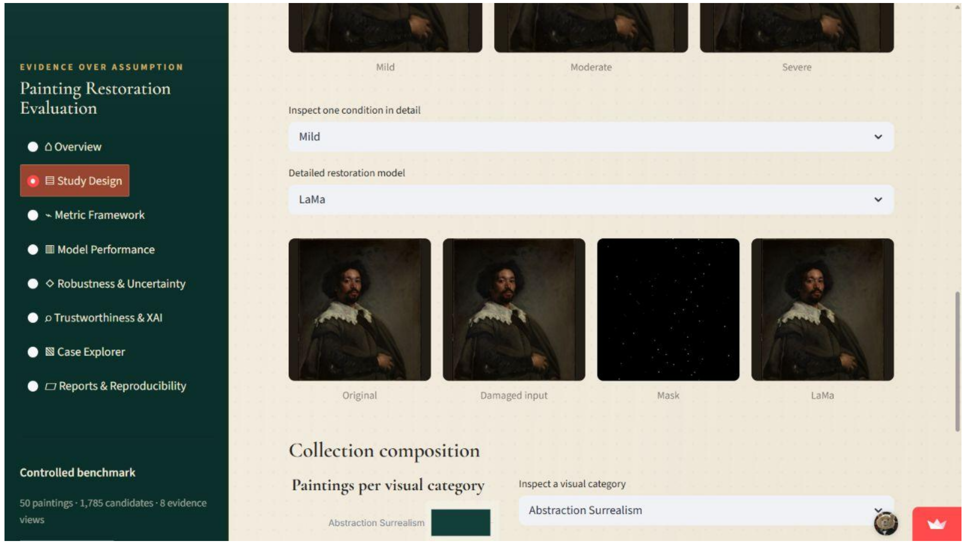
EVIDENCE TRAIL

N07 full degradation registry; N08 eligibility

Show severity progression and eligibility



Mild, moderate, and severe degradation



Eligible masked-removal example and restoration

READ

Water stain, dirt/dust, partial transparency, and water-stain-plus-dirt provide eligible masked-removal cases; other full-image effects remain degradation analyses.

SUPPORTED CONCLUSION

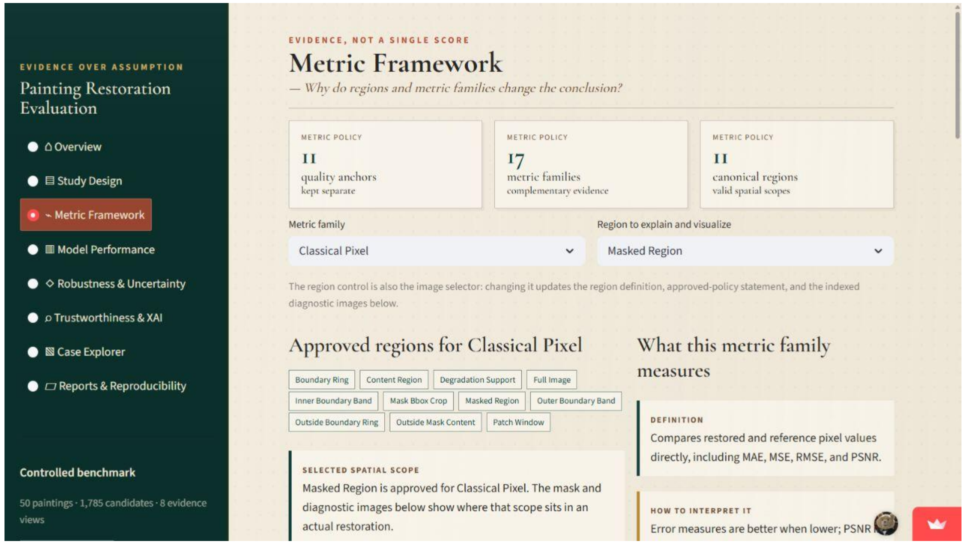
The eligible subset allows a fair inpainting comparison without pretending every degradation is a missing-region task.

LIMIT

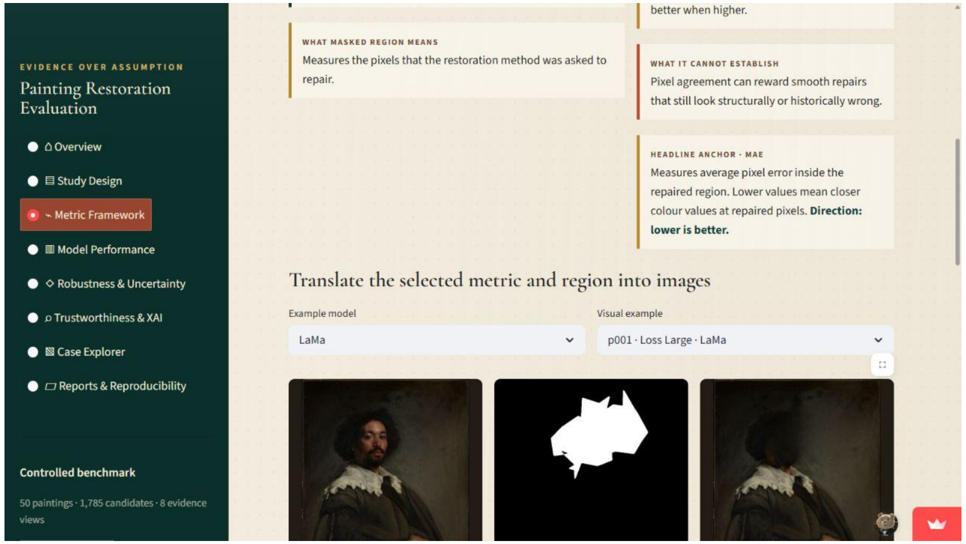
Blur and global colour change are not presented as restoration tasks for these inpainting methods.

Evidence trail: N07 images and audit; N25 sensitivity analysis

Begin with metric family and region together



Classical pixel metrics with region policy



Original, mask, restoration, and selected diagnostic

READ

The metric family answers what is measured; the region answers where. Changing either control changes the definition and the indexed visual explanation.

SUPPORTED CONCLUSION

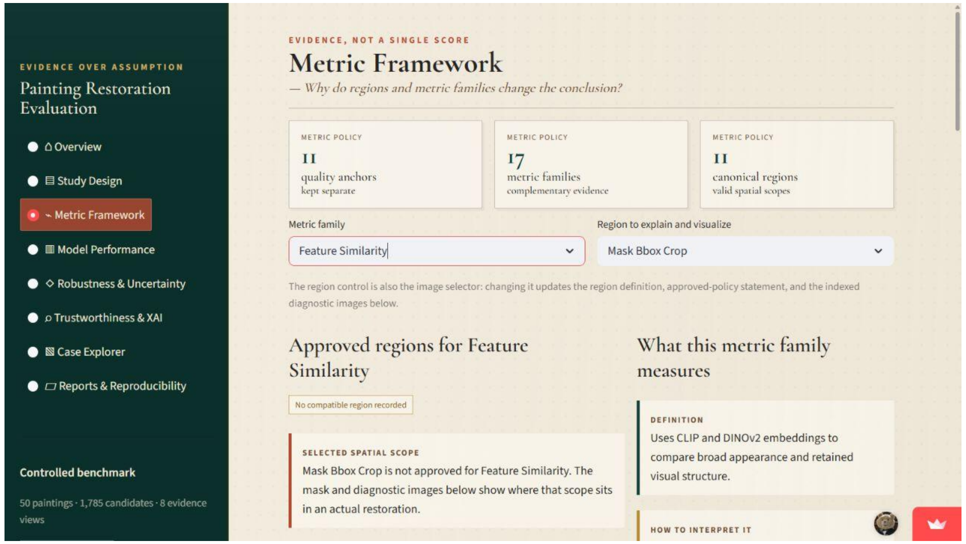
Metric meaning is incomplete without spatial scope.

LIMIT

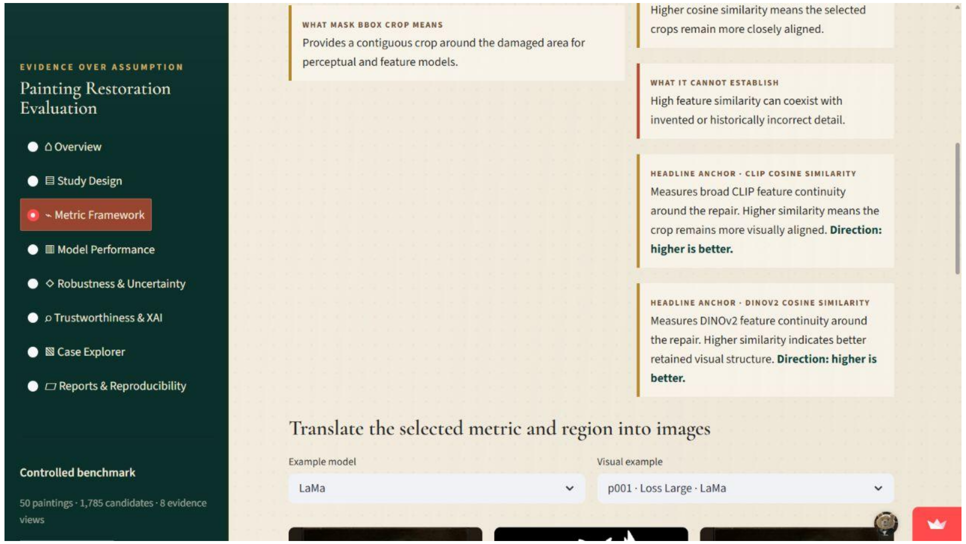
Never compare values across incompatible metrics, regions, or applicability states.

Evidence trail: N08 region policy; N13 classical metrics

Explain feature similarity through contiguous crops



Feature-similarity definition and valid regions



Feature evidence translated into an image crop

READ

CLIP and DINOv2 use the content region or mask-bounding-box crop so the input remains a contiguous image.

SUPPORTED CONCLUSION

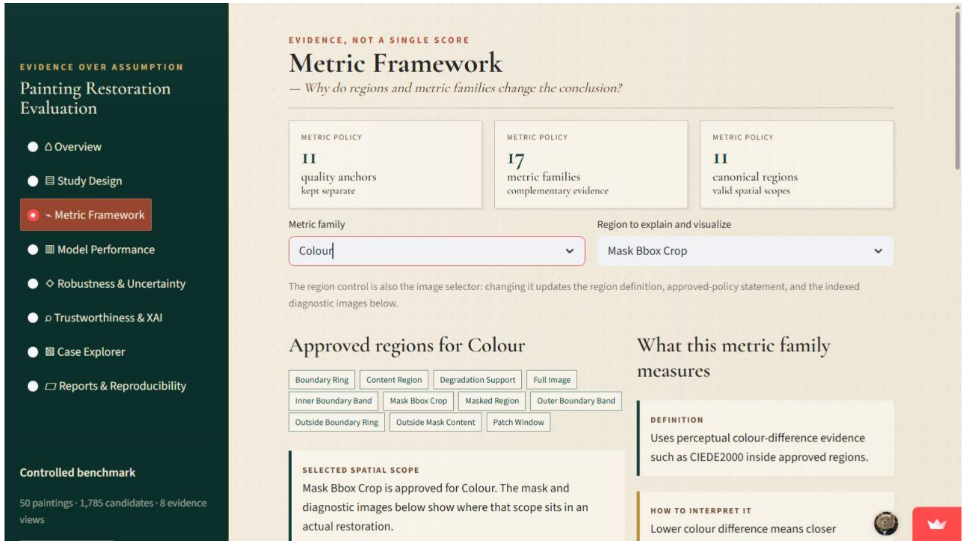
Higher cosine similarity means stronger representation-level alignment with the reference crop.

LIMIT

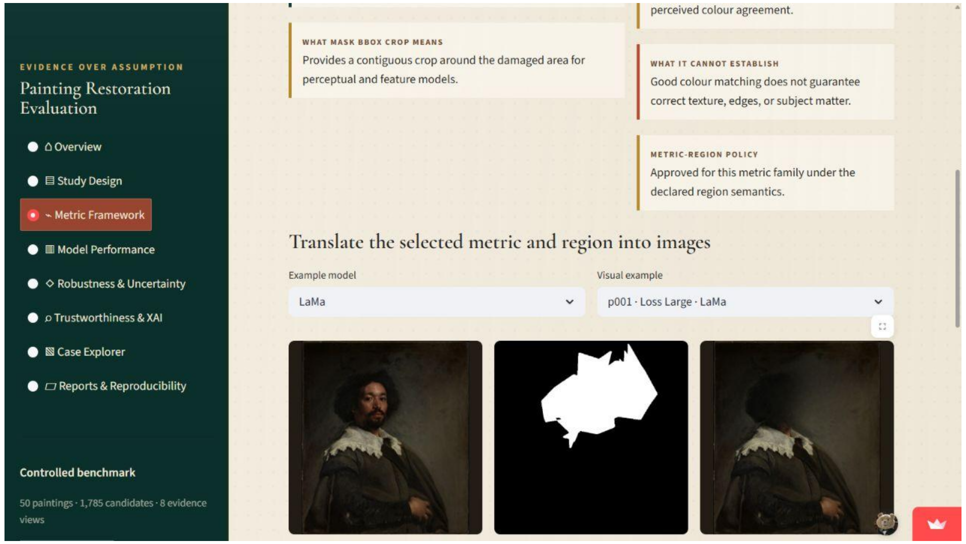
Feature similarity does not prove iconographic accuracy, historical authenticity, or correct fine detail.

Evidence trail: N15 feature_metrics.csv and figures

Read colour evidence as one complementary signal



Colour metric definition



Colour diagnostic for the selected region

READ

CIEDE2000 expresses perceptual colour difference; lower values mean closer perceived colour agreement.

SUPPORTED CONCLUSION

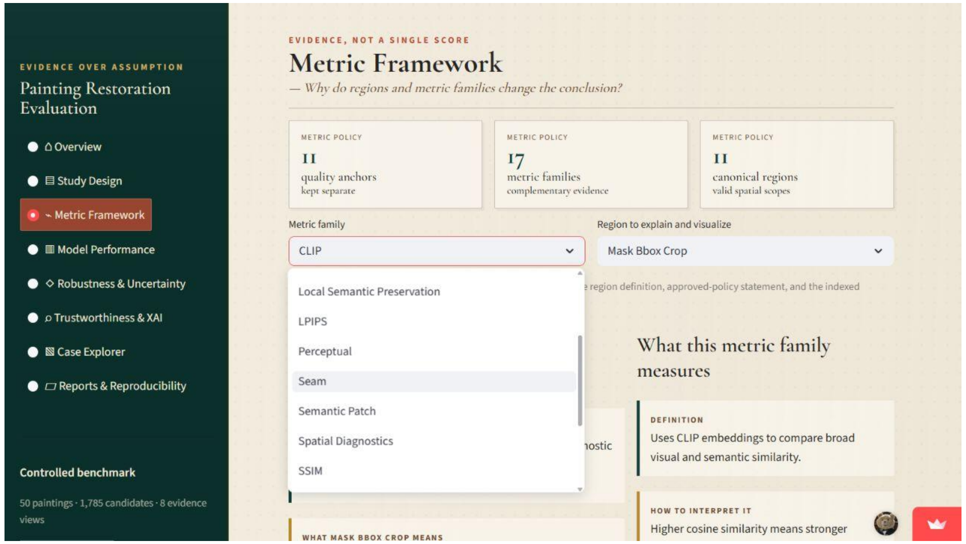
A low colour difference supports colour fidelity in the declared region.

LIMIT

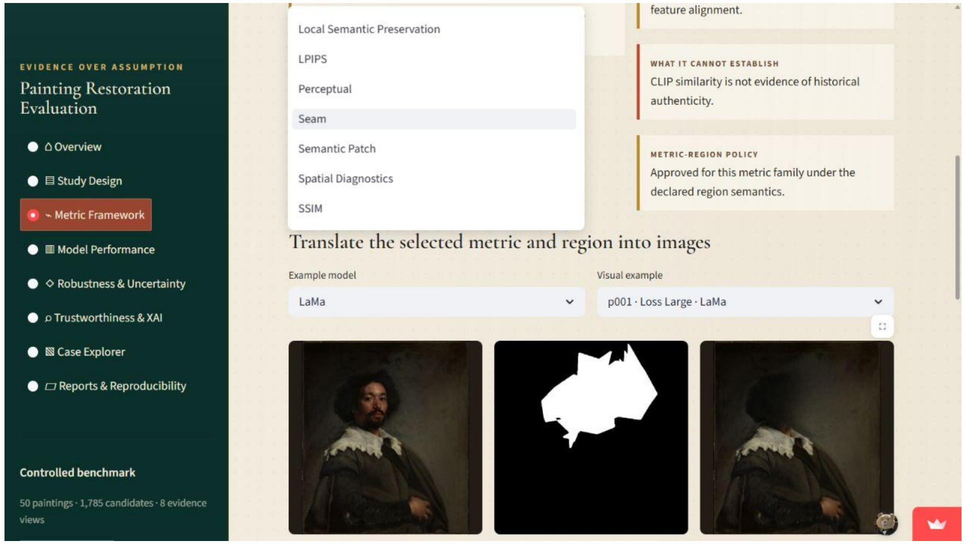
Good colour matching does not establish texture, seam, structure, or semantic correctness.

Evidence trail: N17 local_consistency.csv and colour maps

Inspect the repair boundary explicitly



Seam metric and boundary-ring region



Boundary diagnostic around the repair

READ

Boundary-gradient mismatch examines the ring where generated pixels meet retained original pixels; lower is better.

SUPPORTED CONCLUSION

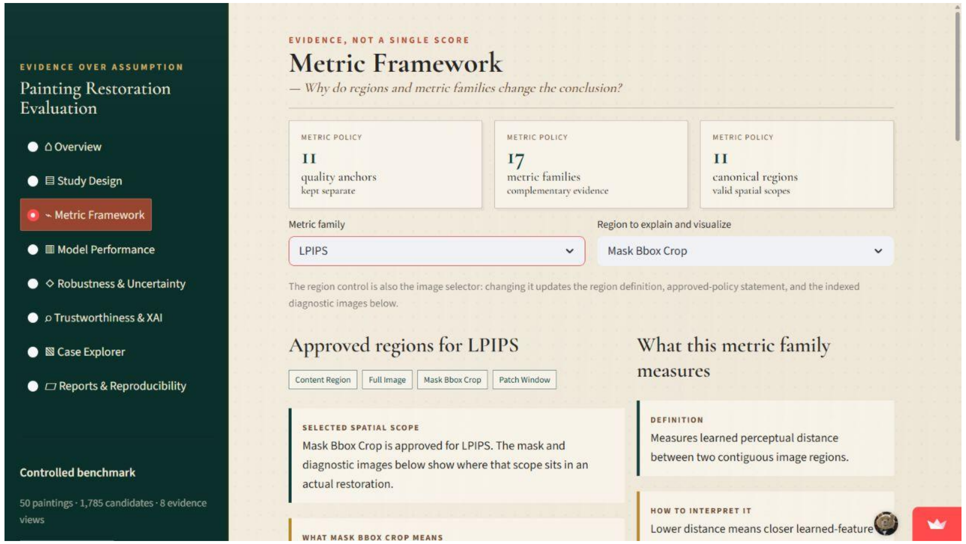
Lower mismatch supports a less visible transition at the repair edge.

LIMIT

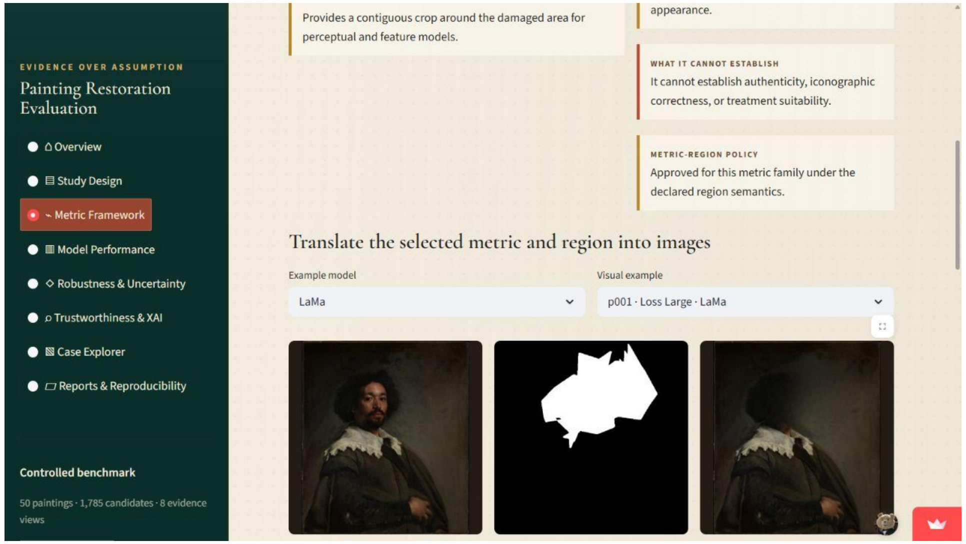
A clean seam can surround an incorrect repair interior.

Evidence trail: N08 boundary-ring policy; N17 seam maps

Use LPIPS for learned perceptual distance



LPIPS definition and approved regions



Perceptual evidence in a contiguous crop

READ

LPIPS compares learned features in contiguous content or mask-bounding-box crops; lower distance means closer learned appearance.

SUPPORTED CONCLUSION

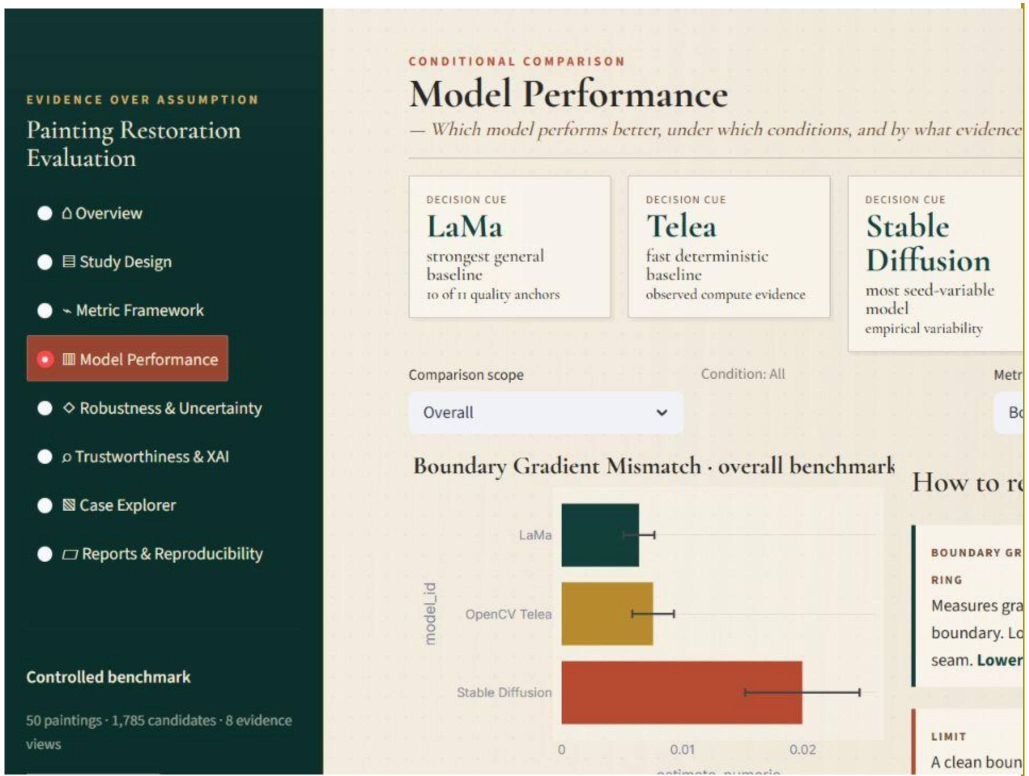
LPIPS adds a perceptual view that exact pixel metrics can miss.

LIMIT

LPIPS remains a learned proxy, not a conservator judgement or authenticity test.

Evidence trail: N14 lpips_metrics.csv

Read every ranking as conditional



Model Performance: one metric, one scope, one grouped estimate

NAVIGATE / SET

Open Model Performance. Choose a comparison scope, condition, and metric.

READ

The plot shows a stored grouped estimate and interval for one declared metric-region anchor.

SUPPORTED CONCLUSION

The displayed winner is valid only for this metric, region, and evidence slice.

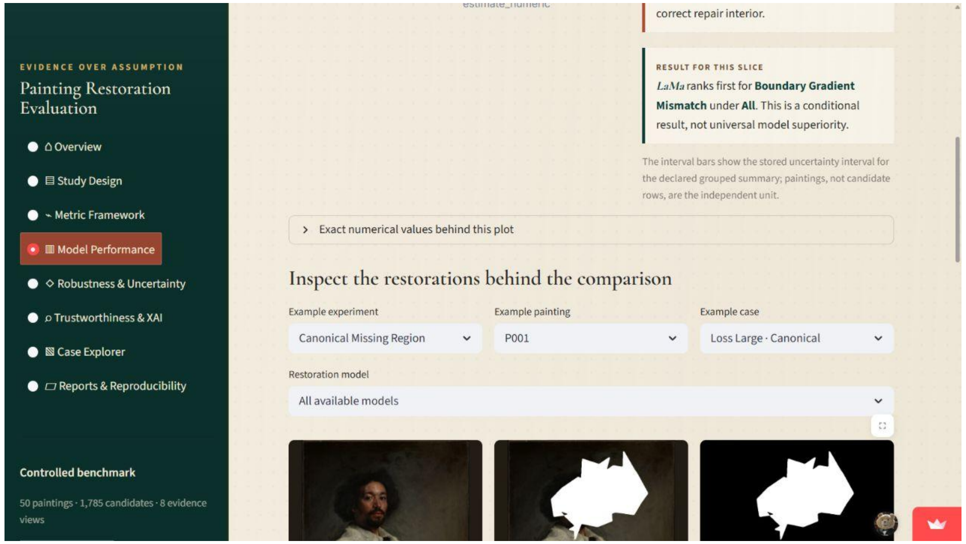
DO NOT CLAIM

Do not turn one metric winner into universal model superiority.

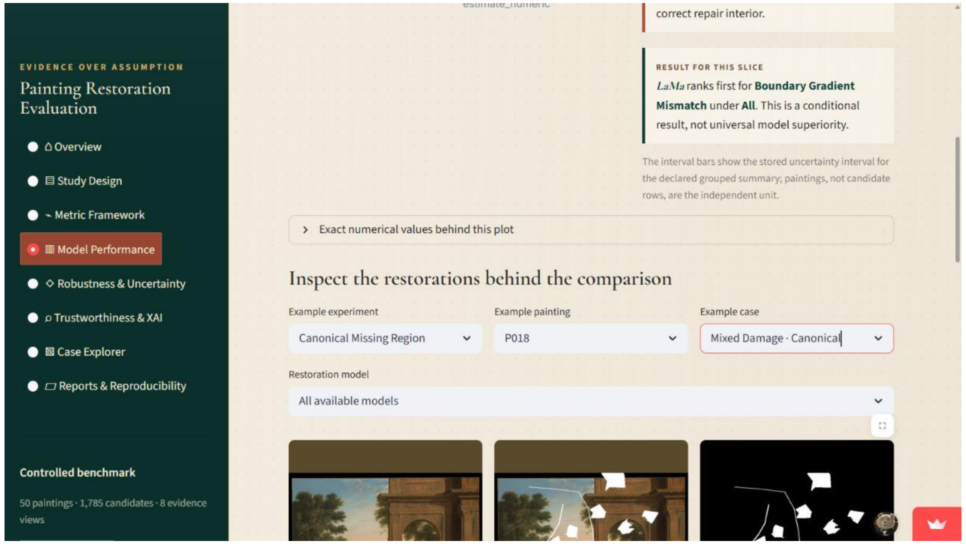
EVIDENCE TRAIL

N21 model_comparison.csv; N26 grouped statistics

Inspect the restorations behind the bar chart



All available models for p001



p018 mixed-damage comparison

READ

The model selector can show all available restorations for a case or isolate one model/seed/prompt arm.

SUPPORTED CONCLUSION

Numerical differences should be checked against the original, damaged input, mask, and restoration images.

LIMIT

A representative image illustrates the metric result; it cannot replace the complete population table.

Evidence trail: N09–N12 images; N34 case index

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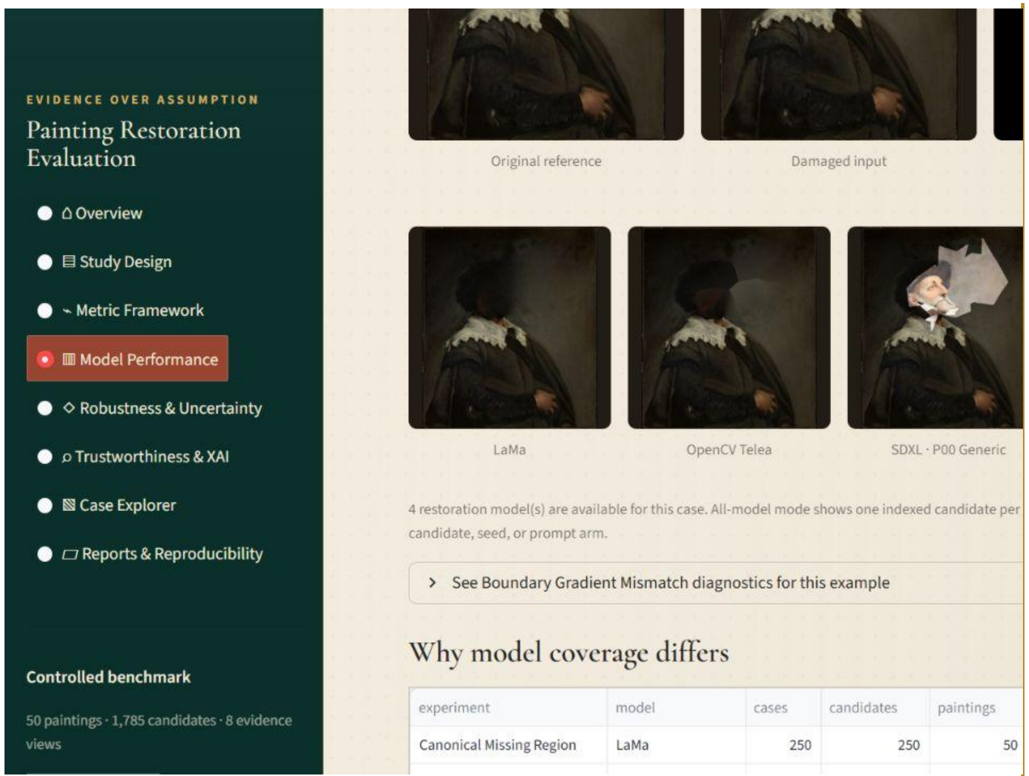
Explain why model coverage differs



Model coverage table and applicability explanation

NAVIGATE / SET

Scroll to 'Why model coverage differs'.

READ

LaMa, Telea, and Stable Diffusion cover all four eligible experiments. SDXL covers four canonical and six synthetic cases only.

SUPPORTED CONCLUSION

SDXL is useful as bounded feasibility evidence but cannot support a fourth full benchmark ranking.

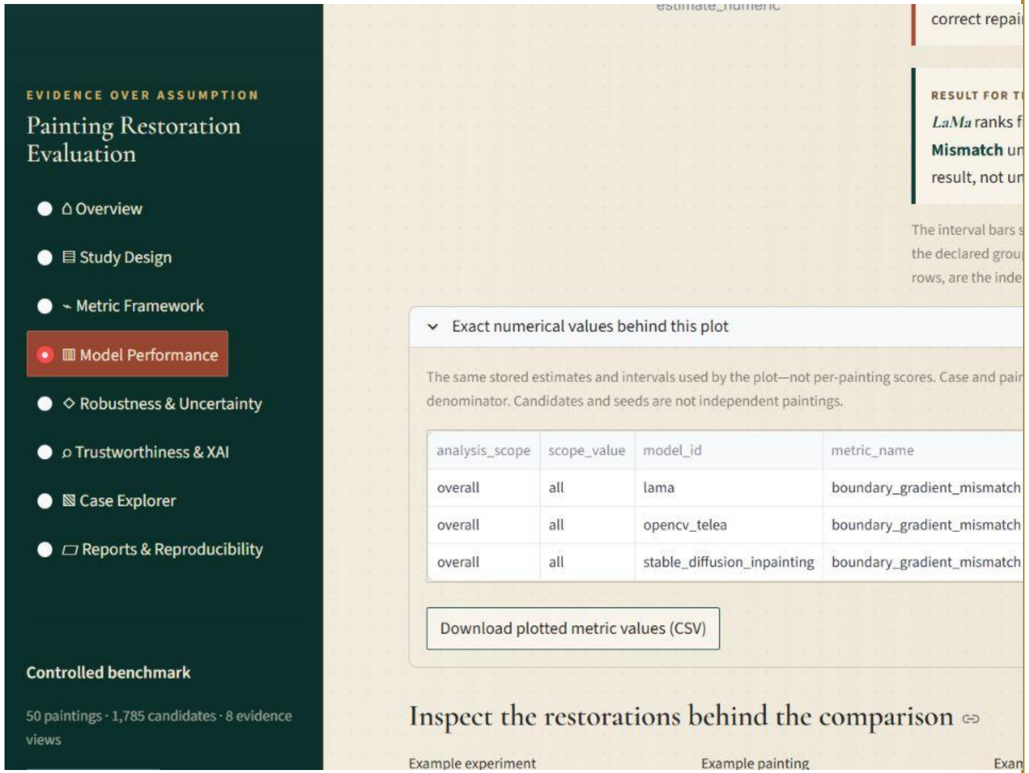
DO NOT CLAIM

Absence means not evaluated under the approved design, not model failure.

EVIDENCE TRAIL

N08 eligibility; N12 SDXL candidates; N34 compute_summary.csv

Open the exact values behind a model plot



Model Performance: expandable numerical table

NAVIGATE / SET

Expand 'Exact numerical values behind this plot'.

READ

The table exposes model, estimate, interval, rank, denominator, region, and direction without recomputation.

SUPPORTED CONCLUSION

The dashboard keeps the numbers behind every visual comparison inspectable.

DO NOT CLAIM

Intervals describe the declared grouped summary; they are not per-candidate uncertainty bars.

EVIDENCE TRAIL

N21 metrics/model_comparison.csv

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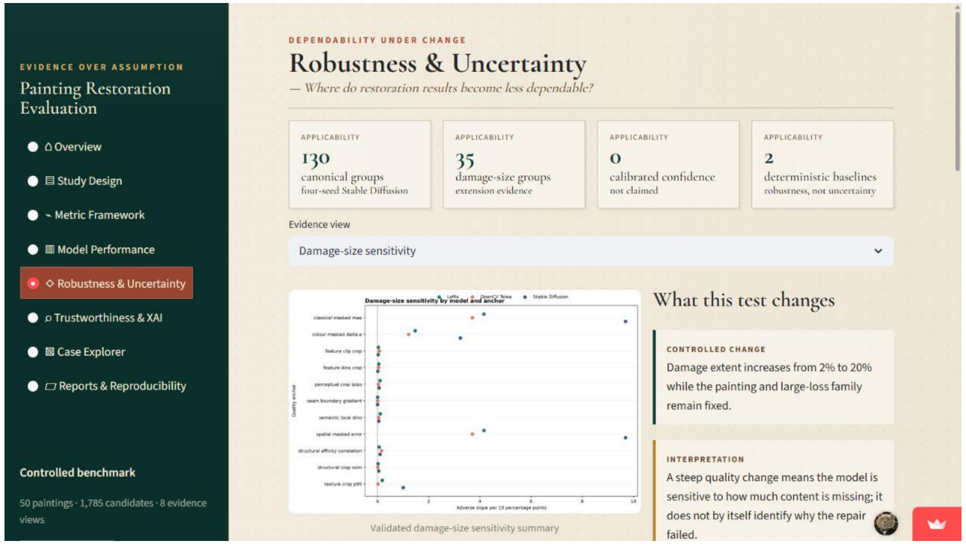
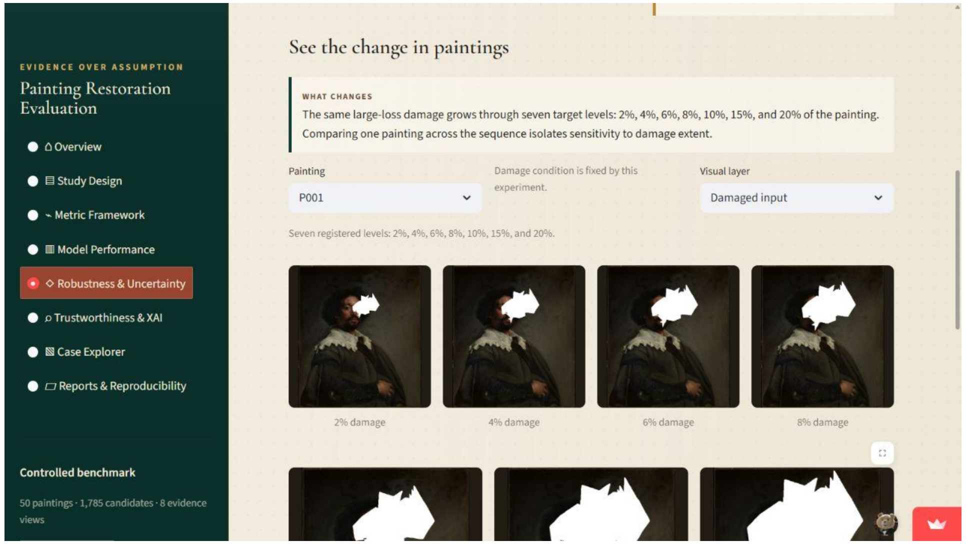
Read a damage-size trajectory



Damage-size sensitivity summary



Visible 2%–20% progression

READ

The same large-loss family grows through seven target levels while painting identity stays fixed.

SUPPORTED CONCLUSION

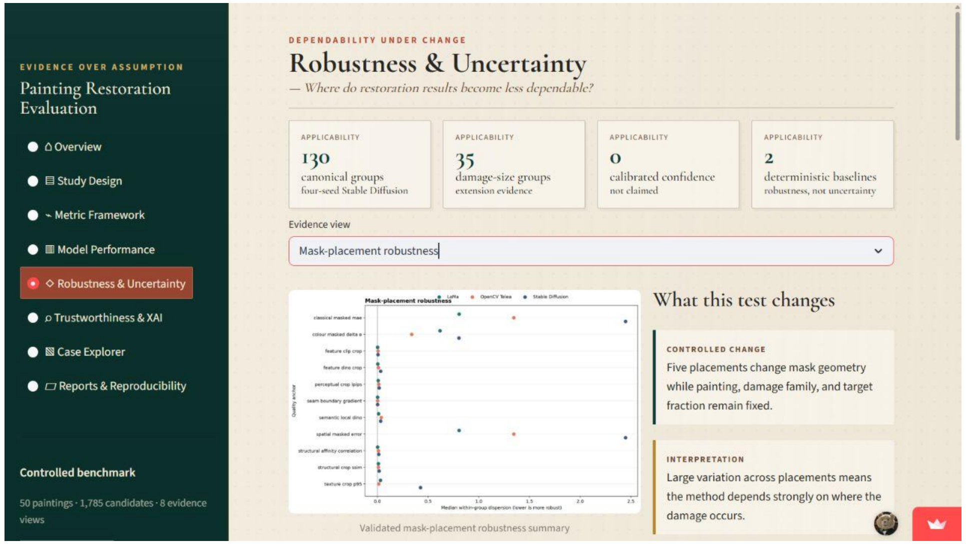
A shallow adverse trajectory indicates greater robustness to increasing missing area for the selected metric.

LIMIT

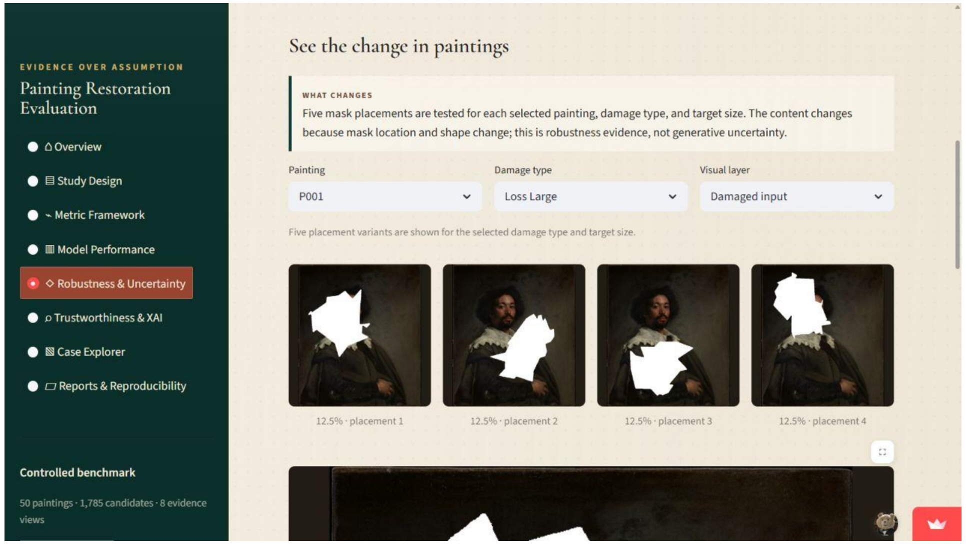
The five-painting cohort does not estimate a population-wide category effect.

Evidence trail: N23 analysis; N05 images

Distinguish mask robustness from uncertainty



Mask-placement grouped evidence



Matched placement examples

READ

Only mask location and geometry change; deterministic and stochastic methods can both be evaluated for this robustness question.

SUPPORTED CONCLUSION

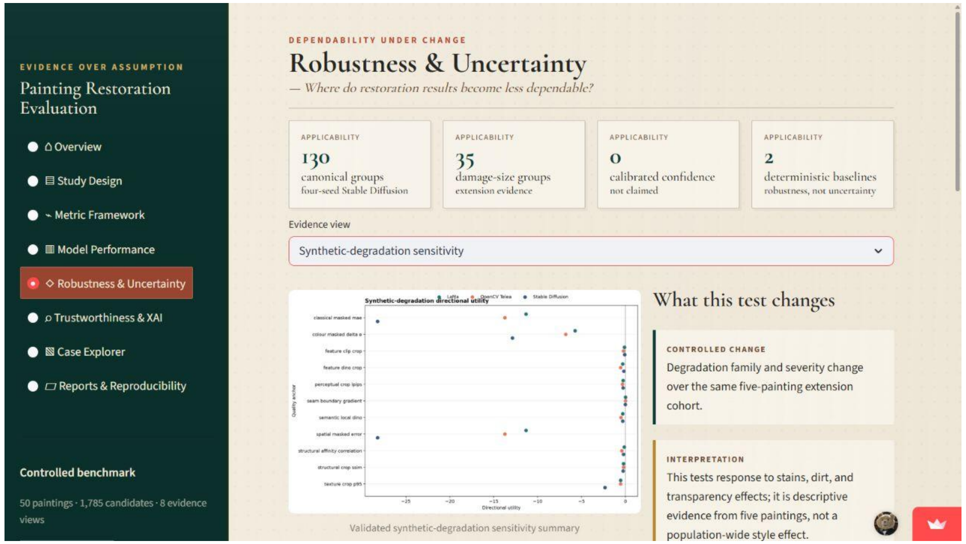
Lower cross-placement variation means greater stability to the controlled mask perturbation.

LIMIT

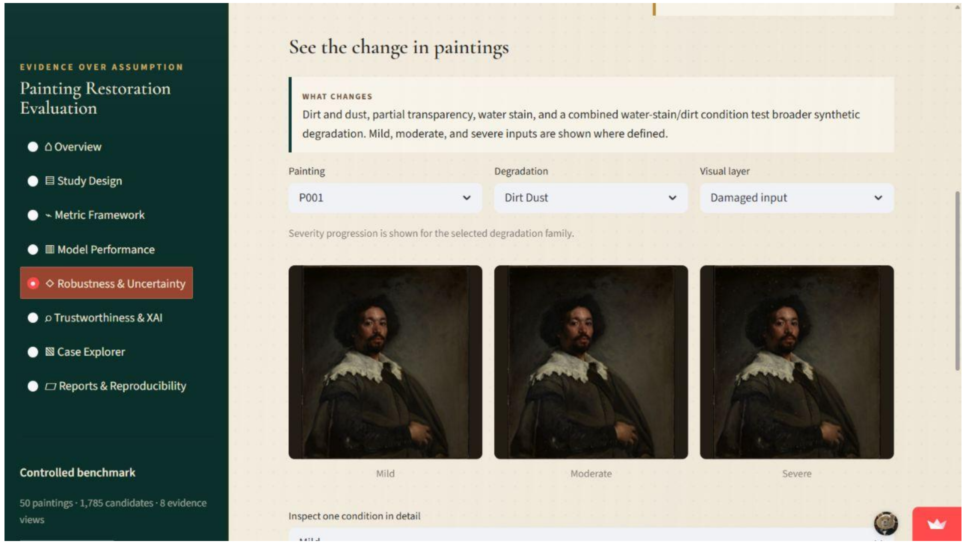
This variation is caused by input geometry, not repeated random seeds.

Evidence trail: N24 analysis; N06 dataset

Read synthetic-degradation sensitivity cautiously



Synthetic-degradation grouped evidence



Severity examples and selected restoration

READ

Compare eligible family/severity combinations while keeping their experiment definition visible.

SUPPORTED CONCLUSION

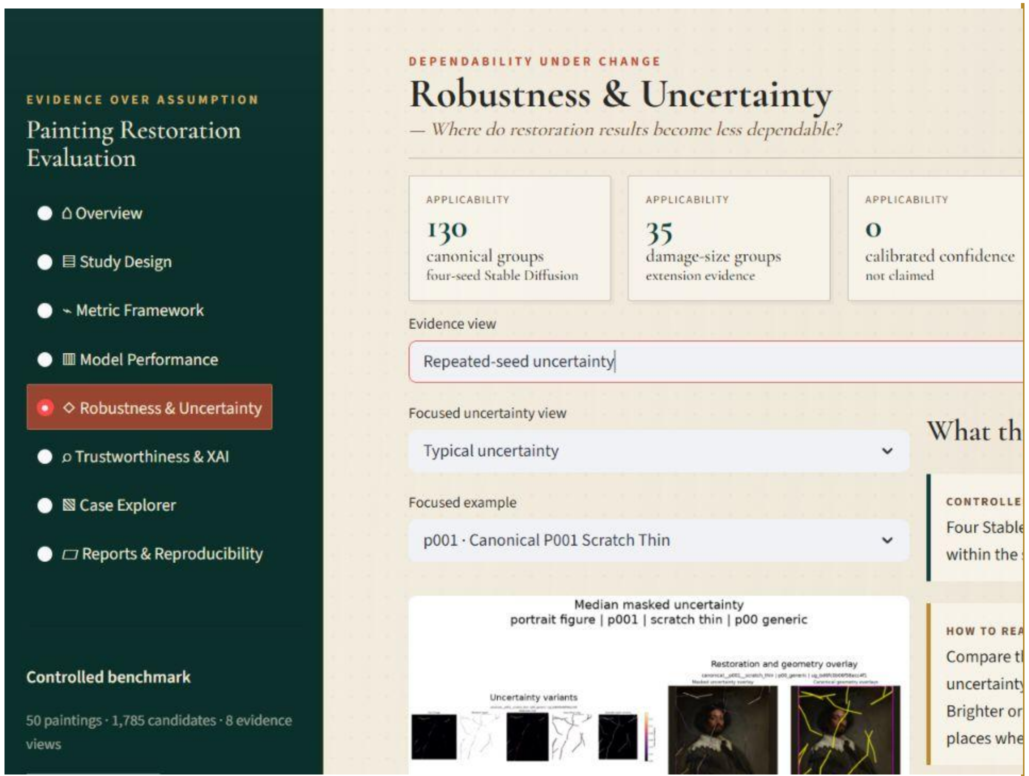
The result reveals which controlled degradation conditions challenge each method more.

LIMIT

Synthetic prevalence and severity do not represent real museum collections.

Evidence trail: N25 analysis; N07 degradation registry

Define repeated-seed uncertainty precisely



Repeated-seed uncertainty: four Stable Diffusion seeds

NAVIGATE / SET

Set Evidence view to Repeated-seed uncertainty.

READ

Each group fixes case, prompt arm, and generation configuration while varying seeds 2026–2029.

SUPPORTED CONCLUSION

Candidate-to-candidate variation measures empirical Stable Diffusion disagreement.

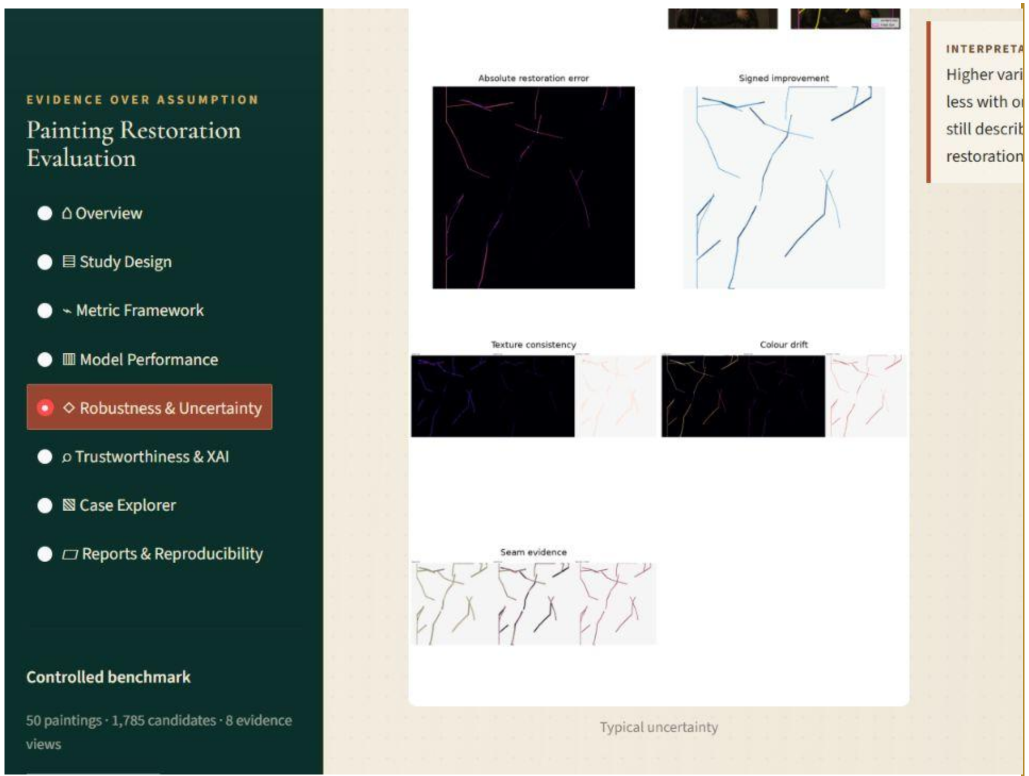
DO NOT CLAIM

It is not calibrated confidence; Telea and LaMa are deterministic, and SDXL has one seed per case.

EVIDENCE TRAIL

N18 canonical groups; N22 damage-size groups

Compare candidates before reading the heatmap



Four seed candidates with mapped diagnostics

NAVIGATE / SET

Choose one uncertainty group; inspect the four restorations first.

READ

Differences among seeds establish whether variation is visible and whether it aligns with the masked or boundary regions.

SUPPORTED CONCLUSION

The map is most useful when it localizes a visible candidate disagreement.

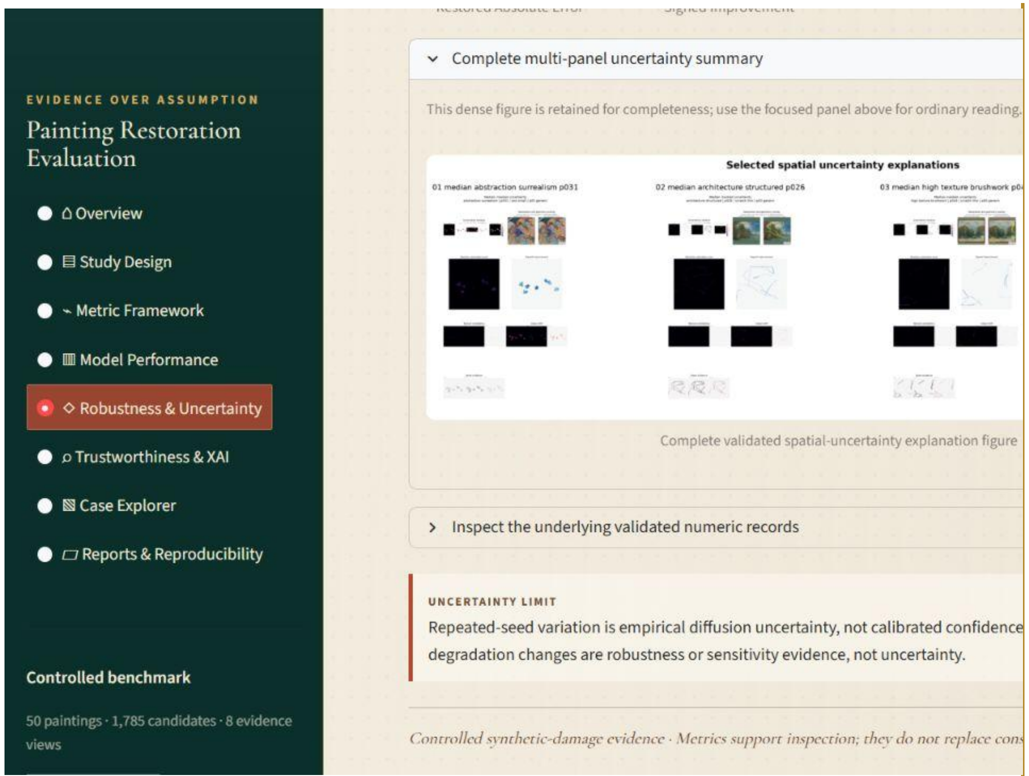
DO NOT CLAIM

A dark map can still hide four consistently incorrect candidates.

EVIDENCE TRAIL

N19 spatial explanation maps; N18 scalar and pairwise uncertainty

Use the multi-panel map as a locator



Expanded uncertainty and spatial-diagnostic panel

NAVIGATE / SET

Expand 'Complete multi-panel uncertainty summary'.

READ

Pixelwise variability, perceptual/feature distances, boundary evidence, and improvement maps answer different questions.

SUPPORTED CONCLUSION

Multiple spatial views help distinguish disagreement inside the repair from boundary or outside-mask effects.

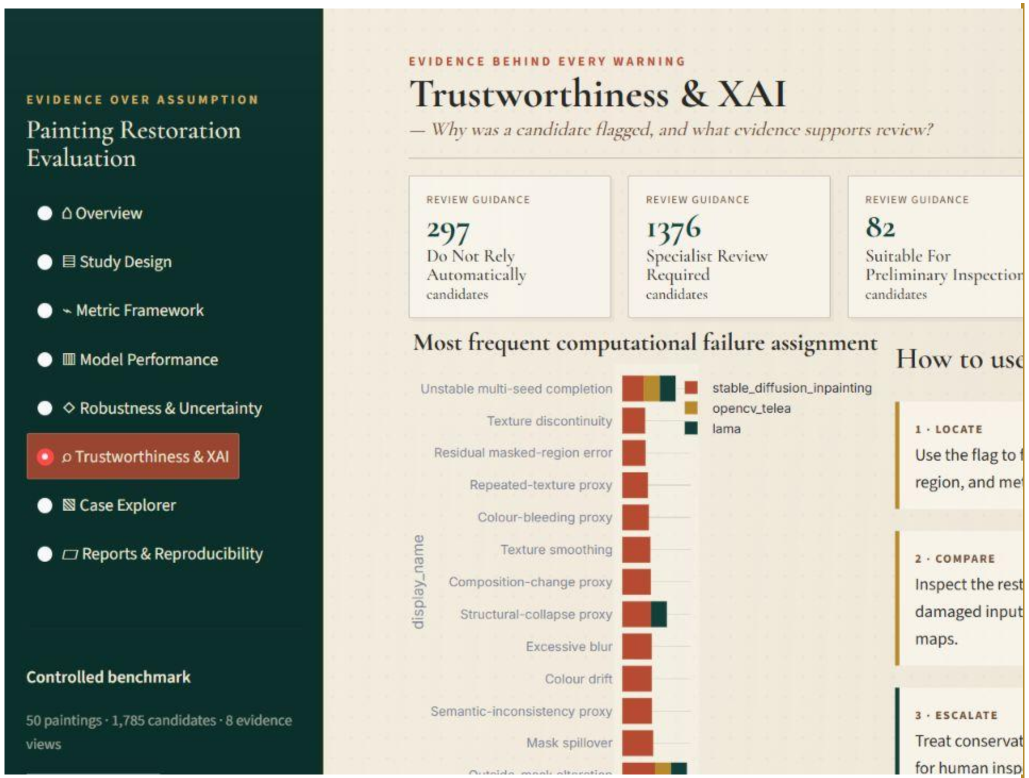
DO NOT CLAIM

No combined uncertainty index is constructed because weights and calibration are not justified.

EVIDENCE TRAIL

N18–N20; N22 extension

Start with the population-wide review lanes



Trustworthiness: conservative review outcomes and failure frequencies

NAVIGATE / SET

Open Trustworthiness & XAI with All outcomes and All models.

READ

297 candidates are 'Do Not Rely Automatically', 1,376 require specialist review, 82 are suitable for preliminary inspection, and 30 are unstable.

SUPPORTED CONCLUSION

1,703 of 1,785 candidates trigger conservative review guidance; human inspection remains central.

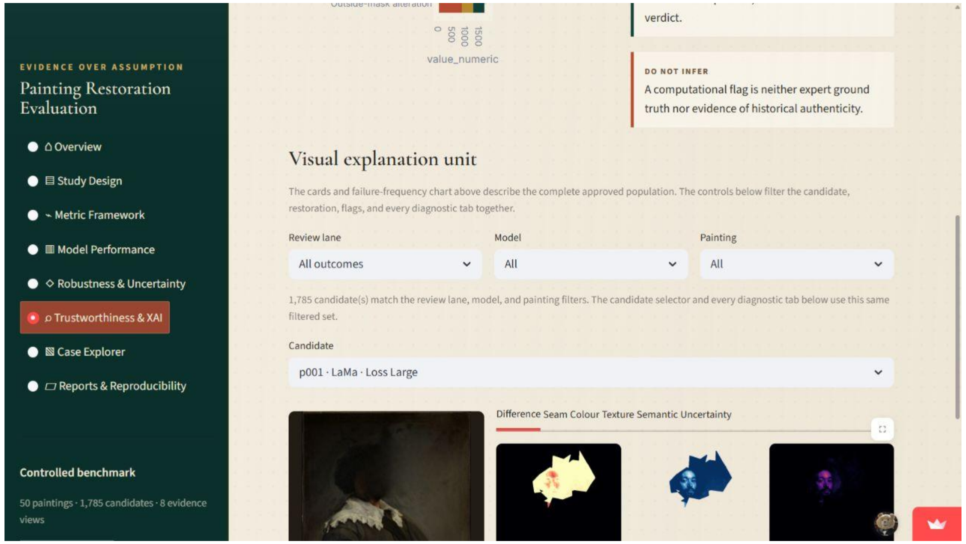
DO NOT CLAIM

The rules are operational computational guidance, not expert labels or conservation approval.

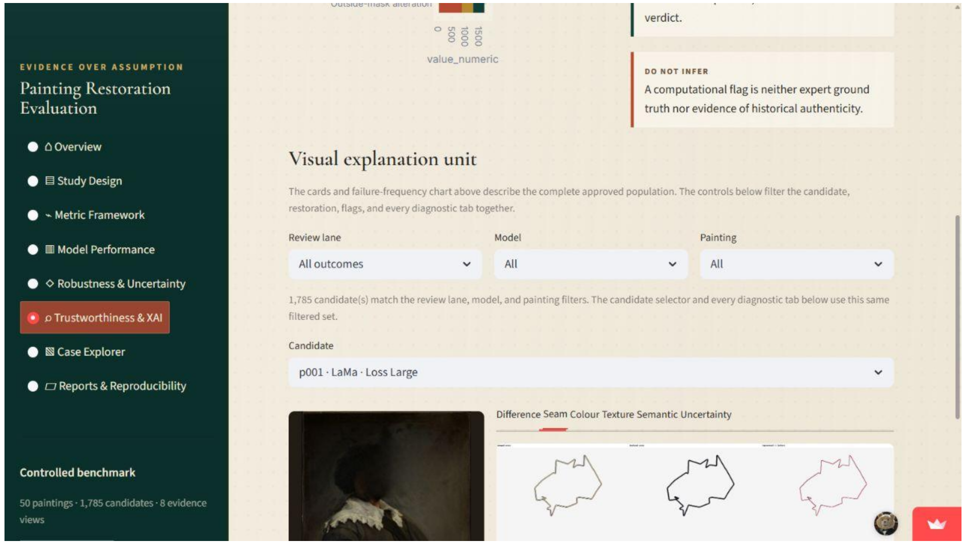
EVIDENCE TRAIL

N27 flags; N29 explanation cases

Follow a flag from reason to visual evidence



One synchronized explanation unit



Boundary and seam diagnostic tab

READ

Filters update the candidate, restoration, flags, and every diagnostic tab together. Use Locate → Compare → Escalate.

SUPPORTED CONCLUSION

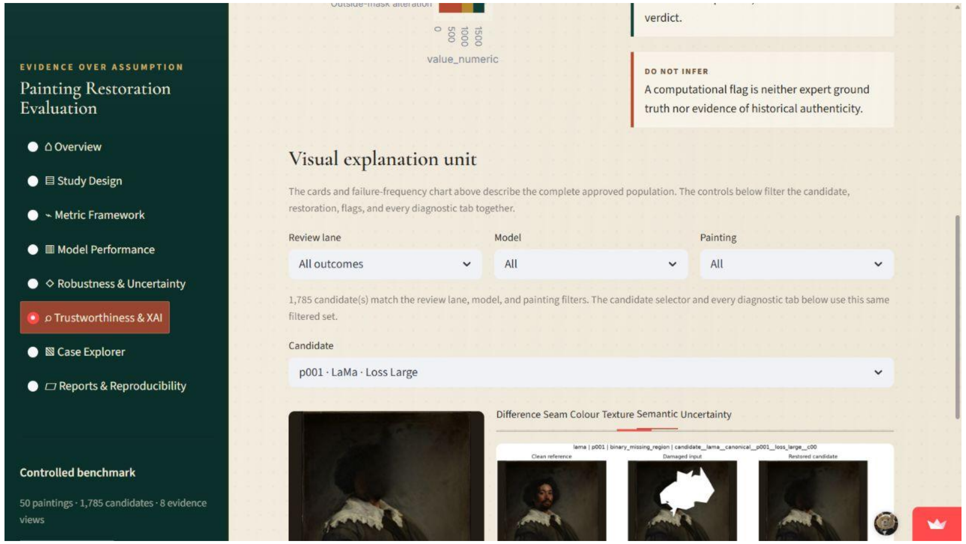
A warning is explainable when the triggering metric, affected region, map, and recommended action remain linked.

LIMIT

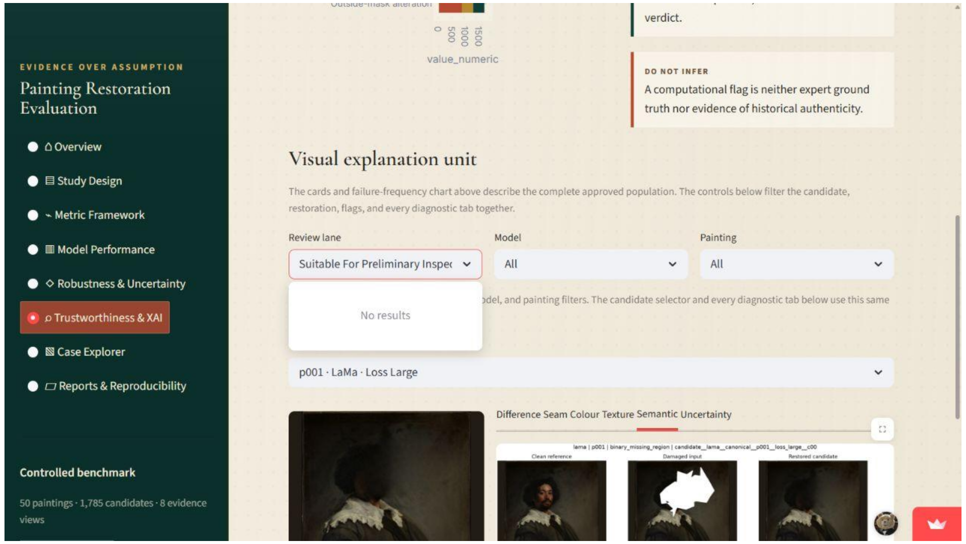
A rule-derived flag is not proof that the restoration is wrong.

Evidence trail: N27 trustworthiness_flags.csv; N29 explanation_cases.csv

Compare semantic evidence and a lower-risk lane



Semantic/structural diagnostic



Suitable-for-preliminary-inspection filter

READ

Semantic maps and feature proxies add context that pixel or seam evidence alone cannot provide. The lower-risk lane is still an inspection lane.

SUPPORTED CONCLUSION

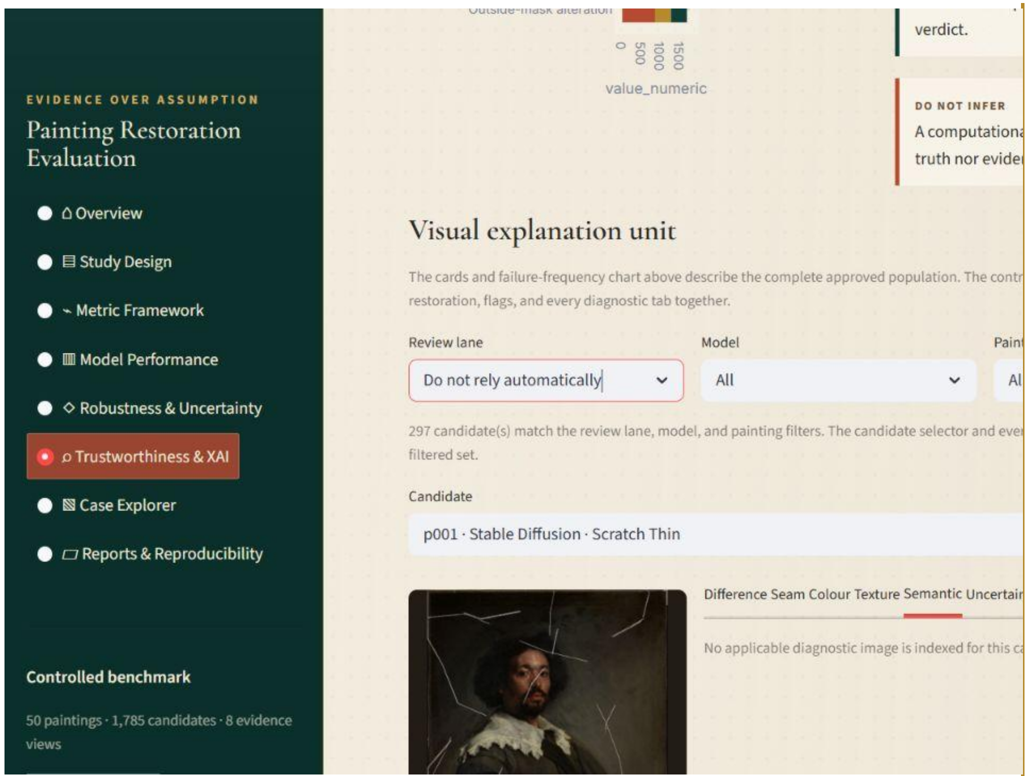
A candidate can be prioritized for preliminary inspection when no stronger operational warning fires.

LIMIT

Suitable for preliminary inspection does not mean historically correct, authentic, or treatment-ready.

Evidence trail: N20 semantic maps; N27 rule assignments

Use the strongest review lane as an escalation cue



Do-not-rely-automatically candidate filter

NAVIGATE / SET

Set Review lane to Do Not Rely Automatically; optionally filter by model or painting.

READ

The candidate selector is restricted to the filtered population, and the recommended actions explain what to inspect next.

SUPPORTED CONCLUSION

The dashboard identifies candidates that should not be accepted without explicit human review.

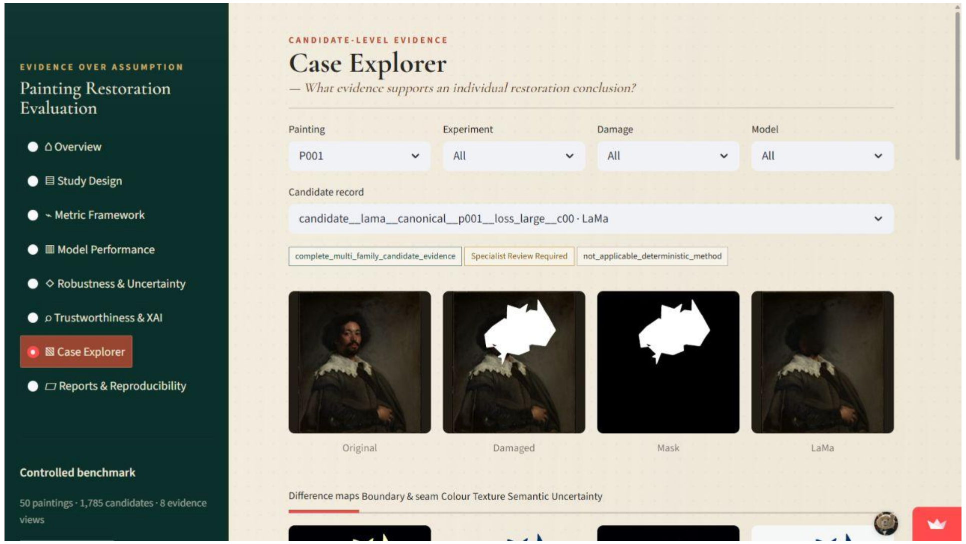
DO NOT CLAIM

It does not decide the restoration's conservation suitability.

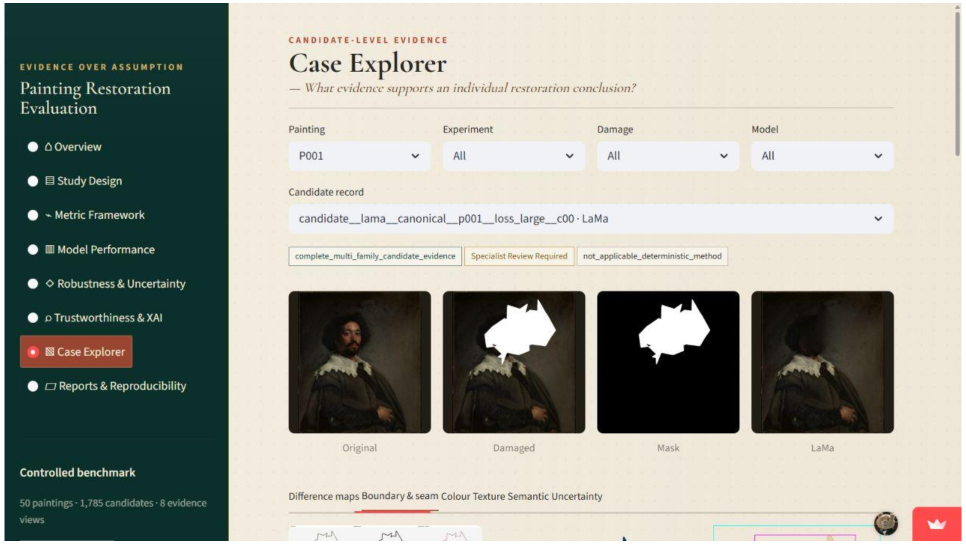
EVIDENCE TRAIL

N27 recommendation categories; N29 counterfactual explanations

Read one candidate from image to boundary map



p001 candidate and primary images



p001 boundary/seam diagnostics

READ

The top row fixes painting, experiment, damage, model, and candidate identity. Diagnostic tabs reveal distinct evidence without changing the candidate.

SUPPORTED CONCLUSION

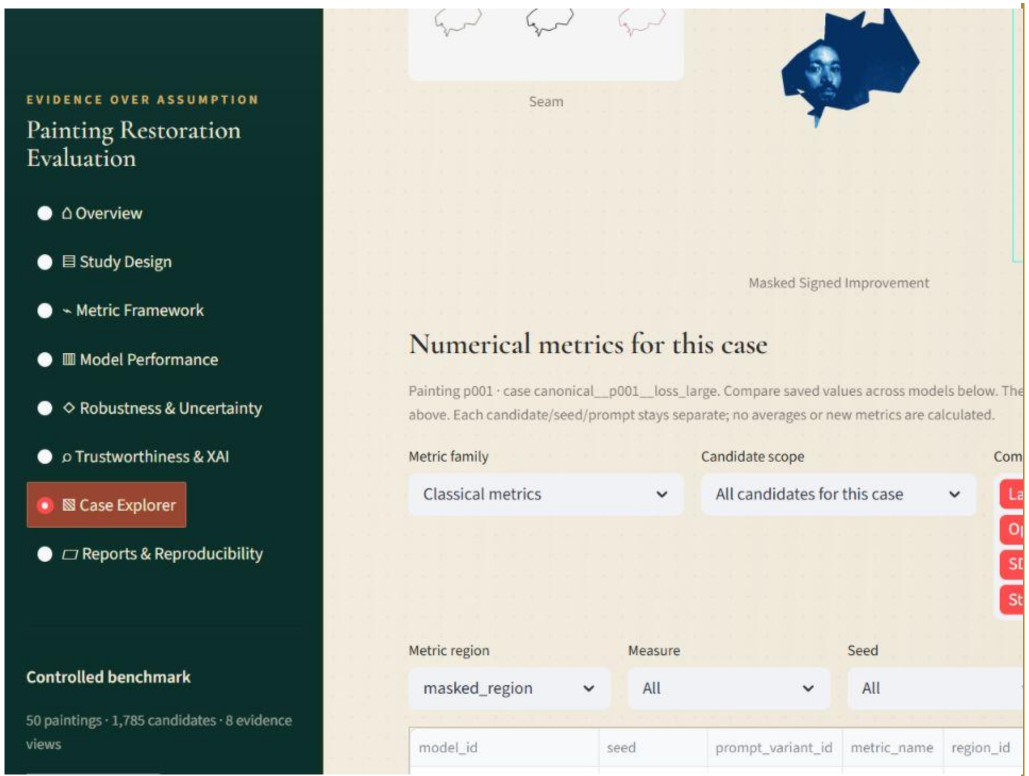
A case conclusion is defensible when image, metric, diagnostic, recommendation, and provenance refer to the same candidate ID.

LIMIT

Do not mix evidence from different seeds, prompt arms, cases, or models.

Evidence trail: N34 case_index.csv; N16/N17 maps

Inspect exact numerical metrics for a case



Case Explorer: candidate-level numerical table

NAVIGATE / SET

Scroll to Numerical metrics; choose a metric family and candidate scope.

READ

Values remain separated by candidate, metric, region, direction, seed, and prompt arm. Missing or inapplicable values remain explicit.

SUPPORTED CONCLUSION

The report examples are selective, but the dashboard exposes every approved candidate's available numerical evidence.

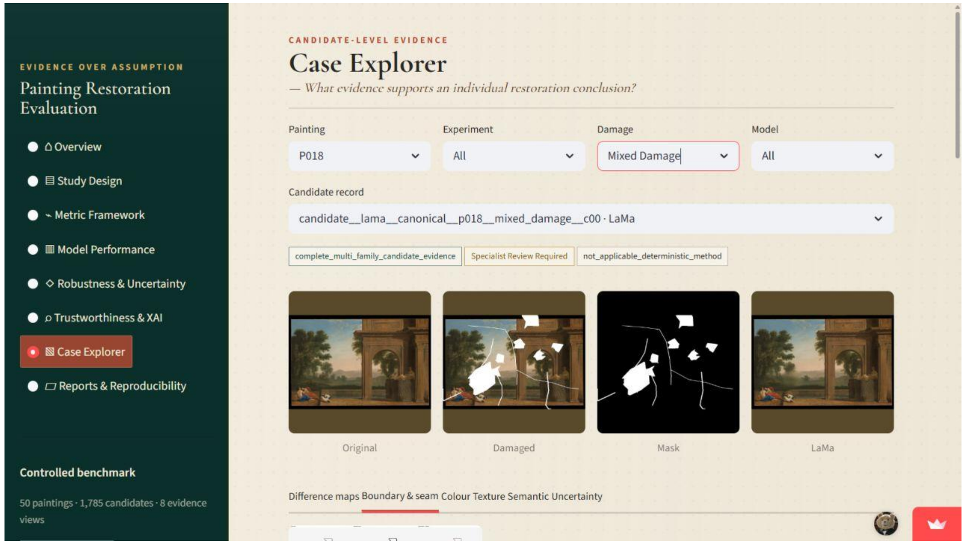
DO NOT CLAIM

The controls do not calculate new averages or fill missing evidence.

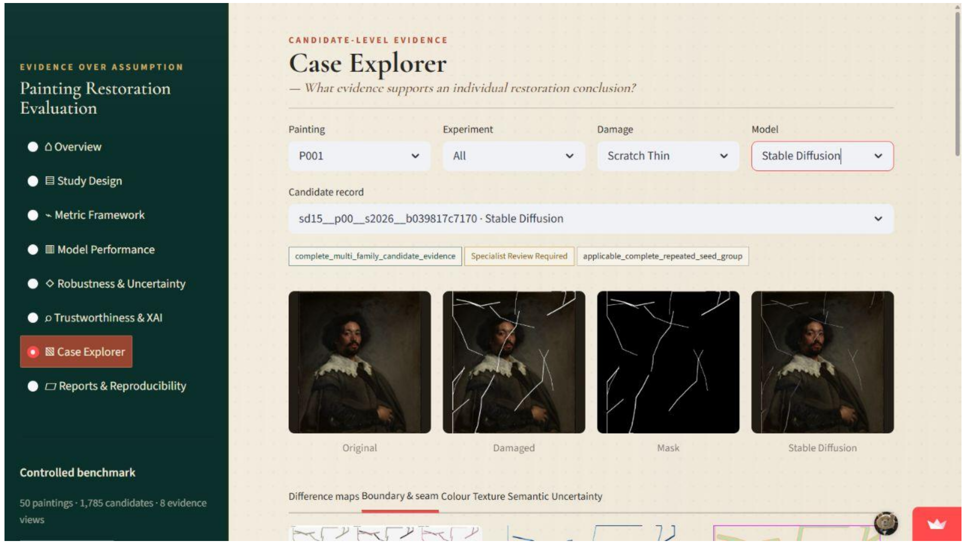
EVIDENCE TRAIL

N13–N20 producer tables; N34 approved allow-list

Use p018 and p001 for contrasting stories



p018 mixed damage



p001 thin-scratch Stable Diffusion

READ

p018 stresses mixed geometry and landscape continuity. p001 thin scratches expose Stable Diffusion's tendency to preserve line-like artifacts under narrow masks.

SUPPORTED CONCLUSION

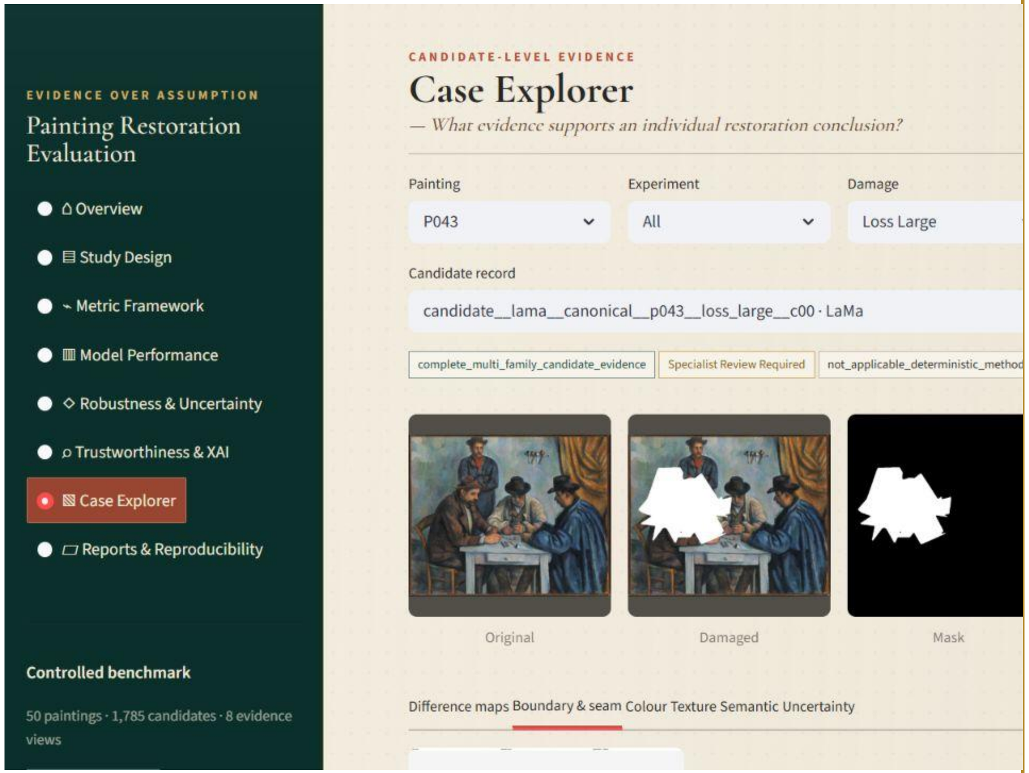
Model suitability depends on damage geometry as well as overall benchmark strength.

LIMIT

A damage-aware prompt is an ablation result, not proof that prompting solves thin-mask geometry.

Evidence trail: N11 scratch prompt ablation; N21 comparison; N32 case reports

Use p043 to discuss texture fidelity



p043 high-texture/brushwork case

NAVIGATE / SET

Set Painting to p043 and inspect texture, seam, and semantic tabs together.

READ

High-texture surfaces make smoothing, repeated texture, and brushwork discontinuity easier to see than a whole-image score suggests.

SUPPORTED CONCLUSION

Texture-aware evidence is necessary because plausible colour and structure can coexist with surface-detail loss.

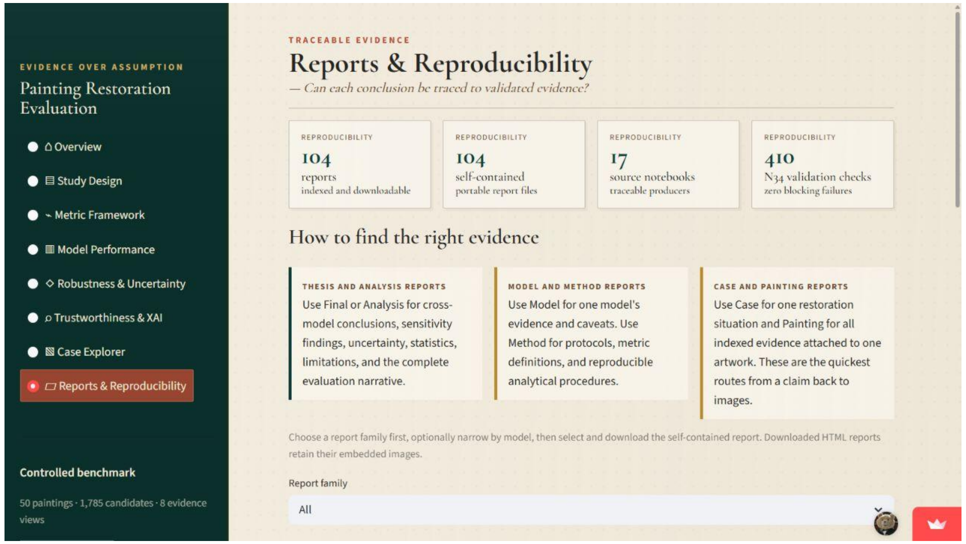
DO NOT CLAIM

Texture descriptors are proxies and do not validate an artist's original technique.

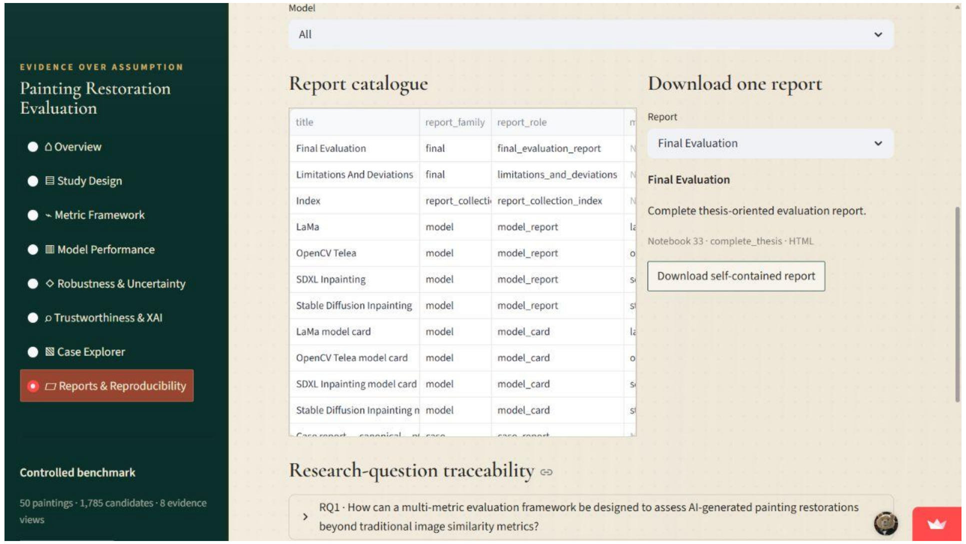
EVIDENCE TRAIL

N17 texture maps; N20 structural evidence

Choose the report family that matches the question



Report-finding guide



Filterable 104-report catalogue

READ

Final/analysis reports support cross-model claims; model/method reports explain methods; case/painting reports return to specific images.

SUPPORTED CONCLUSION

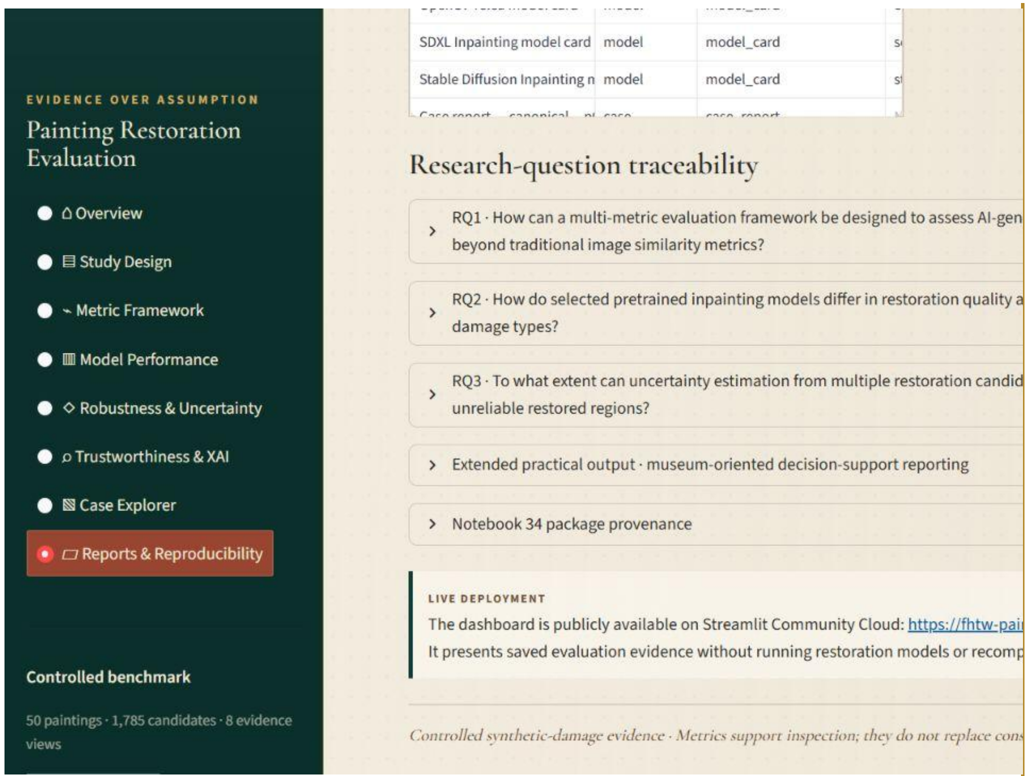
The report catalogue provides a fast route from a question to the correct level of evidence.

LIMIT

A report summary is not a substitute for checking its denominators, scope, and limitations.

Evidence trail: N31–N33 reports; N34 report_index.csv

Download a self-contained report



Report selection and portable download

NAVIGATE / SET

Filter the family/model, select a report, and use Download self-contained report.

READ

All 104 indexed HTML reports retain embedded images when downloaded independently.

SUPPORTED CONCLUSION

Supervisor and reviewer evidence can be shared without depending on local image paths.

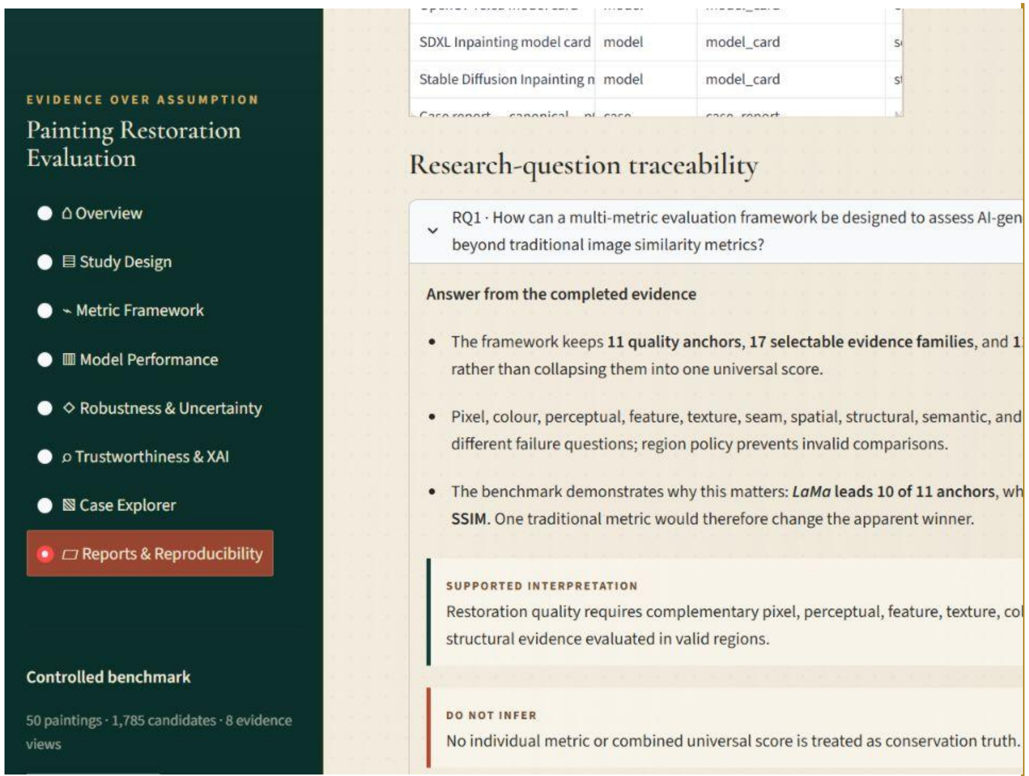
DO NOT CLAIM

Downloaded reports contain saved evidence only; they do not rerun models or metrics.

EVIDENCE TRAIL

N31 model reports; N32 case/painting reports; N33 final reports

Answer each research question through its evidence chain



Research-question traceability and live deployment

NAVIGATE / SET

Expand RQ1, RQ2, and RQ3 in Research-question traceability.

READ

Each question lists the evidence role, producing notebooks, source tables, and limitations; the live deployment link is shown below.

SUPPORTED CONCLUSION

The dashboard is an index into a reproducible evidence system, not a standalone collection of screenshots.

DO NOT CLAIM

Traceability proves provenance and validation status, not the truth of a conservation judgement.

EVIDENCE TRAIL

N34 research_question_coverage.csv; N36 supervisor package

Metric family quick reference

Metric/evidence	Measures	Preferred direction	Valid spatial idea	Key limit
MAE / MSE / RMSE	Exact pixel error	Lower	Masked, content, full, boundary and other approved pixel regions	Can reward smooth but structurally wrong repairs
PSNR	Pixel reconstruction ratio	Higher	Same approved classical regions	Scale-dependent and visually incomplete
SSIM	Local luminance, contrast, structure	Higher	Contiguous regions such as bbox/content	May disagree with colour, texture and seam
LPIPS	Learned perceptual distance	Lower	Content region; mask-bounding-box crop	Learned proxy, not expert judgement
CLIP cosine	Broad visual/semantic feature alignment	Higher	Content region; bbox crop; patch where approved	Not historical authenticity
DINOv2 cosine	Representation-level visual structure	Higher	Content region; bbox crop	Does not prove generated detail
CIEDE2000	Perceptual colour difference	Lower	Masked and other approved colour regions	Colour alone cannot validate structure
Texture descriptors/maps	Local surface and directional continuity	Closer / lower error	Bbox, boundary, content as declared	Compresses complex brushwork
Boundary-gradient mismatch	Transition at repair edge	Lower	Boundary ring	Clean edge can surround wrong interior
Spatial diagnostics	Where error or improvement occurs	Lower error / stronger improvement	Map-specific declared regions	A scalar can hide spatial concentration
Semantic/structural proxies	Local feature and layout retention	Higher similarity/correlation	Contiguous patches or content	Cannot verify iconography
Repeated-seed variability	Candidate disagreement	Context-dependent	Six scalar regions; contiguous learned regions	Not confidence or correctness

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Canonical region dictionary

Region ID	What it contains	Why it exists
full_image	All 768×768 pixels, including padding	Whole-frame checks; use cautiously when padding/content differ
content_region	Recorded painting content bounds after aspect-preserving padding	Whole-painting evidence without padded border
masked_region	Pixels explicitly requested for repair	Direct repair accuracy, colour, local error
mask_bbox_crop	Contiguous crop enclosing the mask	SSIM, LPIPS, CLIP, DINOv2 and local context
inner_boundary_band	Thin band just inside the mask edge	Generated-side transition
outer_boundary_band	Thin band just outside the mask edge	Original-side transition and spillover
boundary_ring	Combined inner and outer bands	Seam and boundary continuity
outside_mask_content	Painting content excluding the repair mask	Context preservation
outside_boundary_ring	Content outside both mask and boundary ring	Unintended distant alteration
degradation_support	Pixels affected by a procedural degradation	Synthetic-degradation localization
patch_window	Declared local contiguous patch	Local semantic and structural feature evidence

Why contiguous regions matter: SSIM, LPIPS, CLIP, and DINOv2 operate on image structure or learned image features. Arbitrary sparse masked pixels are not a valid visual input for those models; the mask-bounding-box crop preserves spatial layout.

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The 11 overall quality anchors

Anchor	Region	Better	LaMa	Telea	Stable Diffusion
mae	masked region	Lower	14.5370 (rank 1)	17.2579 (rank 2)	28.9894 (rank 3)
delta e ciede2000 mean	masked region	Lower	6.9310 (rank 1)	7.8580 (rank 2)	12.9566 (rank 3)
clip cosine similarity	mask bbox crop	Higher	0.9565 (rank 1)	0.8863 (rank 3)	0.9431 (rank 2)
dinov2 cosine similarity	mask bbox crop	Higher	0.8716 (rank 1)	0.6966 (rank 3)	0.8478 (rank 2)
local patch cosine similarity	mask bbox crop	Higher	0.7999 (rank 1)	0.6030 (rank 3)	0.6709 (rank 2)
lpips	mask bbox crop	Lower	0.1251 (rank 1)	0.1864 (rank 3)	0.1824 (rank 2)
boundary gradient mismatch	boundary ring	Lower	0.0064 (rank 1)	0.0076 (rank 2)	0.0200 (rank 3)
restored error mean	masked region	Lower	14.5370 (rank 1)	17.2579 (rank 2)	28.9894 (rank 3)
ssim	mask bbox crop	Higher	0.8704 (rank 2)	0.8722 (rank 1)	0.8253 (rank 3)
reference affinity map correlation	content region	Higher	0.9558 (rank 1)	0.8537 (rank 3)	0.8751 (rank 2)
local texture error p95	mask bbox crop	Lower	0.4333 (rank 1)	0.6316 (rank 2)	2.1929 (rank 3)

Reading rule: LaMa ranks first on ten anchors. Telea's narrow SSIM lead is real and worth showing because it demonstrates why a single metric can change the apparent winner.

Population arithmetic and method coverage

Experiment	Case population	LaMa	Telea	Stable Diffusion	SDXL	Design note
Canonical missing region	250 cases	250	250	690	4	50 zero controls + 200 damaged cases
Damage-size sensitivity	35 cases	35	35	140	0	5 paintings × 7 damage levels; four SD seeds
Mask robustness	75 cases	75	75	75	0	5 paintings × matched placements/conditions
Eligible synthetic degradation	50 of 165 registered	50	50	50	6	Masked-removal subset only
Approved totals	410 eligible cases	410	410	955	10	1,785 candidate records

Method	Recorded execution count	Observed workstation runtime	How to explain it
OpenCV Telea	410	185.61 s total	Fast deterministic classical baseline
LaMa	410	655.09 s total	Learned deterministic baseline; strongest general quality result
Stable Diffusion	1,330 executed	12,269.50 s total	Approved catalogue retains 955 candidates; execution count includes additional controlled generation roles
SDXL	10	3,784.69 s total	Bounded feasibility only; runtime is not comparable as a full benchmark

Notebook responsibility map · part 1

N	Notebook	Responsibility group	Notebook-owned output root
01	Dataset Verification	Foundation and experiment contracts	outputs/01_dataset_verification/
02	Image Preprocessing	Foundation and experiment contracts	outputs/02_image_preprocessing/
03	Canonical Mask Generation	Foundation and experiment contracts	outputs/03_canonical_mask_generation/
04	Canonical Damaged Image Generation	Foundation and experiment contracts	outputs/04_canonical_damaged_image_generation/
05	Damage Size Sensitivity Dataset Generation	Foundation and experiment contracts	outputs/05_damage_size_sensitivity_dataset_generation/
06	Mask Robustness Dataset Generation	Foundation and experiment contracts	outputs/06_mask_robustness_dataset_generation/
07	Synthetic Degradation Dataset Generation	Foundation and experiment contracts	outputs/07_synthetic_degradation_dataset_generation/
08	Experiment Contracts And Region Policy	Foundation and experiment contracts	outputs/08_experiment_contracts_and_region_policy/
09	Opencv Telea Restoration	Restoration inference	outputs/09_opencv_telea_restoration/
10	Lama Restoration	Restoration inference	outputs/10_lama_restoration/
11	Stable Diffusion Restoration	Restoration inference	outputs/11_stable_diffusion_restoration/
12	Sdxl Feasibility Or Restoration	Restoration inference	outputs/12_sdxl_feasibility_or_restoration/
13	Classical Metrics	Metrics, maps, semantics, uncertainty	outputs/13_classical_metrics/
14	Lpips Metrics	Metrics, maps, semantics, uncertainty	outputs/14_lpips_metrics/
15	Feature Similarity	Metrics, maps, semantics, uncertainty	outputs/15_feature_similarity/
16	Difference Maps And Spatial Diagnostics	Metrics, maps, semantics, uncertainty	outputs/16_difference_maps_and_spatial_diagnostics/
17	Local Consistency Metrics	Metrics, maps, semantics, uncertainty	outputs/17_local_consistency_metrics/
18	Diffusion Uncertainty Analysis	Metrics, maps, semantics, uncertainty	outputs/18_diffusion_uncertainty_analysis/

Notebook responsibility map · part 2

N	Notebook	Responsibility group	Notebook-owned output root
19	Uncertainty And Spatial Explanation Maps	Metrics, maps, semantics, uncertainty	outputs/19_uncertainty_and_spatial_explanation_maps/
20	Semantic And Structural Consistency	Metrics, maps, semantics, uncertainty	outputs/20_semantic_and_structural_consistency/
21	Multi Model Comparison	Metrics, maps, semantics, uncertainty	outputs/21_multi_model_comparison/
22	Damage Size Diffusion Uncertainty Extension	Metrics, maps, semantics, uncertainty	outputs/22_damage_size_diffusion_uncertainty_extension/
23	Damage Size Sensitivity Analysis	Sensitivity, statistics, trustworthiness, XAI	outputs/23_damage_size_sensitivity_analysis/
24	Mask Robustness Analysis	Sensitivity, statistics, trustworthiness, XAI	outputs/24_mask_robustness_analysis/
25	Synthetic Degradation Analysis	Sensitivity, statistics, trustworthiness, XAI	outputs/25_synthetic_degradation_analysis/
26	Grouped And Statistical Analysis	Sensitivity, statistics, trustworthiness, XAI	outputs/26_grouped_and_statistical_analysis/
27	Failure Taxonomy And Trustworthiness Flags	Sensitivity, statistics, trustworthiness, XAI	outputs/27_failure_taxonomy_and_trustworthiness_flags/
28	Metric And Region Policy Ablation	Sensitivity, statistics, trustworthiness, XAI	outputs/28_metric_and_region_policy_ablation/
29	Explainable Ai And Case Retrieval	Sensitivity, statistics, trustworthiness, XAI	outputs/29_explainable_ai_and_case_retrieval/
30	Model Cards Compute And Scalability	Model cards and report generation	outputs/30_model_cards_compute_and_scalability/
31	Model Report Generation	Model cards and report generation	outputs/31_model_report_generation/
32	Case And Painting Report Generation	Model cards and report generation	outputs/32_case_and_painting_report_generation/
33	Final Evaluation Report	Model cards and report generation	outputs/33_final_evaluation_report/
34	Final Streamlit Dashboard Assets	Dashboard, deployment, supervisor package	outputs/34_final_streamlit_dashboard_assets/
35	Dashboard And Deployment Validation	Dashboard, deployment, supervisor package	outputs/35_dashboard_and_deployment_validation/
36	Supervisor Publication Reproducibility Package	Dashboard, deployment, supervisor package	outputs/36_supervisor_publication_reproducibility_package/

Supervisor questions: short, defensible answers

Question	Answer to give
Why not use one score?	Metric families and regions disagree in meaningful ways. A weighted score would hide those disagreements and require unjustified calibration.
Why call categories broad visual categories?	They provide balanced operational grouping, but they are not independently validated art-historical styles.
Why include a clean reference?	The work is a controlled reconstruction benchmark. The reference allows direct measurement, but it is not historical ground truth for genuinely damaged paintings.
Why does RQ2 refer to inpainting methods rather than pretrained models?	The evaluated set intentionally includes Telea as a classical deterministic baseline alongside learned deterministic and stochastic methods.
Why does LaMa lead?	It ranks first on ten of eleven declared overall anchors across error, colour, perceptual, feature, seam, texture, and structural evidence.
Why mention Telea's SSIM win?	Because it demonstrates metric disagreement: crop SSIM alone gives a different winner and would overstate Telea's overall position.
Why not rank SDXL fully?	Only ten single-seed cases were run. That supports feasibility and runtime evidence, not a fourth full benchmark.
Does high uncertainty mean wrong?	No. Higher repeated-candidate disagreement means lower stochastic stability. It prioritizes inspection; it does not establish error or calibrate correctness.
Are the trustworthiness labels expert judgements?	No. They are transparent, conservative rules derived from computational evidence and linked to recommended review actions.
Can someone inspect every case?	Yes. The dashboard indexes all 1,785 approved candidates, all available numerical evidence, 23,964 visual records, and 104 downloadable reports.

Closing checklist for the supervisor meeting

- Open the live dashboard before the meeting and keep the Case Explorer on p018 mixed damage as the main visual example.
- State the central thesis first: visual plausibility is not the same as restoration trustworthiness.
- Give the scope arithmetic once: 50 paintings, 525 registered cases, 410 restoration-eligible cases, and 1,785 approved candidates.
- Show LaMa's 10-of-11 anchor result, then immediately show Telea's SSIM win to prove that metric disagreement is preserved.
- Use the Metric Framework region selector to explain why masked pixels, bounding-box crops, boundary rings, and surrounding content answer different questions.
- Use p001 scratch-thin Stable Diffusion to discuss narrow-mask limitations and the damage-aware prompt ablation.
- Use the four-seed uncertainty page to distinguish disagreement from correctness and confidence.
- Use Trustworthiness & XAI to demonstrate how a warning links to images, metrics, maps, and an action.
- End in Reports & Reproducibility by expanding RQ1–RQ3 and downloading one self-contained report.
- If challenged on a claim, move down the evidence chain: dashboard summary → exact table → candidate images/maps → producer notebook/manifests.

The dashboard supports transparent inspection. It does not automate conservation judgement.

https://fhtw-painting-restoration.streamlit.app/